

SPRINGER SERIES IN STATISTICS

Equivariant Statistical Estimation with Applications

A Unified Theory of Orbital Spaces,
Twin Kernels, and Group-Theoretic Methods

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*To those who seek the hidden symmetries
in the apparent chaos of data.*

Preface

This book presents a unified theory of statistical estimation that systematically exploits the symmetry structure of data and parameter spaces. The central insight is simple yet powerful: *when data possess inherent symmetries, statistical procedures should respect and exploit these symmetries*.

Classical statistics treats observations as points in Euclidean space, ignoring any underlying structure. This approach has two fundamental limitations: it fails to exploit symmetry for improved efficiency, and it may produce estimators that violate natural invariance properties. We address both issues by developing estimation theory on *orbital spaces*—quotient spaces induced by group actions.

The Three Pillars

Our framework rests on three foundational concepts:

1. **The Twin Metric.** Given a metric space (E, d) with group action G , we define the twin metric on the quotient space:

$$d_G(O(x), O(y)) = \inf_{g \in G} d(x, g \cdot y),$$

which induces the natural topology on E/G .

2. **Orbital Empirical Measures.** Given i.i.d. observations $X_1, \dots, X_n$ from a distribution μ on E , the orbital empirical measure is the pushforward under the quotient map:

$$\hat{\mu}_n^G = \frac{1}{n} \sum_{i=1}^n \delta_{O(X_i)} = \pi_* \hat{\mu}_n.$$

3. **Equivariant Estimators.** Estimators that respect the group action, ensuring geometric coherence and achieving improved statistical efficiency through orbit-averaging.

Organization

The book is organized in three parts:

Part I: Foundations develops the mathematical framework. We introduce orbital spaces, twin metrics, and transported kernels. The key concepts are the twin metric on quotient spaces and the TwinKernel hierarchy of transported smoothing kernels. We establish fundamental properties including the Separation Theorem for orbital function spaces.

Part II: Core Theory presents the main statistical results. We prove orbital versions of the Glivenko–Cantelli and Donsker theorems, develop equivariant kernel density estimation with

variance reduction proportional to orbit size, establish limit theorems in twin geometries, and derive concentration inequalities adapted to multiplicative structures.

Part III: Applications demonstrates the methodology across domains. We apply orbital methods to density estimation, point process intensity estimation, adaptive function approximation, and high-dimensional problems with intrinsic symmetry.

Central Message

The central message of this book is that **symmetry is a statistical resource**. In the presence of group structure, the quotient space is not a technicality—it is the natural home of statistical inference.

Jocelyn Nembé
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Notation and Symbols

Spaces and Sets

$\mathbb{R}, \mathbb{R}^d$	Real line, d -dimensional Euclidean space
$\mathbb{R}_{>0}$	Positive real numbers
$\mathbb{Z}, \mathbb{N}$	Integers, natural numbers
S^1, S^{d-1}	Unit circle, unit sphere in $\mathbb{R}^d$
Δ^{d-1}	$(d-1)$ -dimensional simplex
$L^p(I), L^p(\mu)$	Lebesgue spaces
$H^s(I)$	Sobolev space of order s
$C^{k,\alpha}(I)$	Hölder space
E/G	Quotient space of E by group G
$O(x)$	Orbit of x under group action
$\mathcal{O}_n(\mathcal{H})$	Orbital space at level n with reference $\mathcal{H}$

Group Actions

$G = \langle \phi \rangle$	Cyclic group generated by ϕ
$\mathbb{Z}_m$	Cyclic group of order m
S_n	Symmetric group on n elements
ϕ^n	n -fold composition of ϕ
$\Phi_n = \log^{(n)}$	n -fold iterated logarithm
d_G	Twin metric (orbital metric)
$\pi : E \rightarrow E/G$	Quotient map
ψ	Linearization map
$x \odot y$	Twin operation

Probability and Statistics

$\mathbb{E}, \text{Var}, \text{Cov}$	Expectation, variance, covariance
$\mathbb{P}$	Probability measure
μ, μ^G	Measure on E , pushforward to E/G
$\hat{\mu}_n, \hat{\mu}_n^G$	Empirical measure, orbital empirical measure
$f, \hat{f}_n$	Density, density estimator
$\lambda, \hat{\lambda}_n$	Intensity, intensity estimator
$I(\theta), I^G(\theta)$	Fisher information, orbital Fisher information
$N(t), M(t)$	Counting process, martingale

Kernels and Estimation

K, K_h	Kernel, scaled kernel with bandwidth h
K_j^G	TwinKernel at level j
$\mathcal{N}(\varepsilon, F, d)$	ε -covering number
$N_{[]}(\varepsilon, F, d)$	Bracketing number
MISE, ISE	Mean/Integrated squared error
d_{eff}	Effective dimension

Part I

Foundations

Chapter 1

Introduction: Statistics and Symmetry

1.1 The Role of Symmetry in Statistics

Many statistical problems possess inherent symmetries. When measuring angles, data live on a circle with rotational symmetry. When analyzing compositions, data lie on a simplex with permutation structure. In physics, measurements may be invariant under gauge transformations. In finance, returns are naturally multiplicative rather than additive.

Classical statistical theory typically ignores these symmetries, treating all observations as points in Euclidean space $\mathbb{R}^d$. This approach has fundamental limitations:

- (i) It fails to exploit symmetry structure for improved estimation efficiency.
- (ii) It may produce estimators that violate natural invariance properties.
- (iii) It uses coordinate-dependent methods on intrinsically coordinate-free problems.

This book develops a systematic framework for statistical estimation that respects and exploits symmetry. The key insight is that statistical procedures should operate on *orbits* rather than individual points when the underlying structure exhibits group symmetry.

1.2 Motivating Examples

1.2.1 Directional Data

Consider observations $\theta_1, \dots, \theta_n$ on the unit circle S^1 , representing wind directions, animal migration headings, or circadian rhythms. The data have rotational symmetry: a rotation of all observations by a fixed angle α should not change statistical conclusions.

Classical methods that treat θ_i as points in $[0, 2\pi)$ can produce paradoxical results. For example, the arithmetic mean of $\theta_1 = 1^\circ$ and $\theta_2 = 359^\circ$ is 180° , while the natural directional mean is 0° .

Our framework recognizes that the natural parameter space is the quotient S^1/G for an appropriate symmetry group G , and develops estimators that respect this structure.

1.2.2 Positive Data and Multiplicative Structure

Consider positive observations $X_1, \dots, X_n$ (e.g., stock prices, concentrations, survival times). The natural operations are multiplicative: ratios X_i/X_j and geometric means $(\prod X_i)^{1/n}$ are more meaningful than differences and arithmetic means.

The group $G = (\mathbb{R}_{>0}, \times)$ acts by multiplication. Under the logarithmic isomorphism $\psi(x) = \log x$, this becomes the additive group $(\mathbb{R}, +)$. Statistical procedures should respect this structure.

Our TwinKernel framework provides density estimators, concentration inequalities, and limit theorems adapted to multiplicative geometry.

1.2.3 Compositional Data

When observations represent proportions summing to one, they lie on the simplex $\Delta^{d-1} = \{x \in \mathbb{R}_{>0}^d : \sum_i x_i = 1\}$. If components are exchangeable, the symmetric group S_d acts by permutation, and the natural space is Δ^{d-1}/S_d —the set of unordered compositions.

Our orbital estimation theory shows that exploiting this symmetry reduces the effective dimension and improves efficiency.

1.2.4 Adaptive Function Approximation

Consider approximating an unknown function f from noisy observations. Classical methods fix a basis (polynomials, wavelets, etc.) and approximate within that basis. But different functions require different bases: $\sin(x)$ needs Fourier, $x^{2.5}$ needs power functions.

Our *orbital approximation* framework replaces fixed bases with floating reference spaces adapted via diffeomorphisms. The Separation Theorem guarantees that the optimal deformation level can be identified from data.

1.3 Main Contributions

This book makes five main contributions:

1. **Orbital Empirical Process Theory.** We establish Glivenko–Cantelli and Donsker theorems for empirical measures on quotient spaces E/G , with convergence in the twin metric d_G .
2. **Equivariant Estimation with Variance Reduction.** We show that orbit-averaging reduces variance by a factor of $|G|$ when G is finite—a direct statistical benefit of symmetry.
3. **TwinKernel Density and Intensity Estimation.** We develop kernel methods that transport across group orbits, achieving optimal rates in effective dimension $d_{\text{eff}} = \dim(E) - \dim(G)$.
4. **Limit Theorems in Twin Geometry.** We prove laws of large numbers, central limit theorems, Berry–Esseen bounds adapted to multiplicative and other non-Euclidean structures.
5. **Concentration Inequalities.** We derive Hoeffding, Bernstein, and isoperimetric inequalities that exploit group structure, with dramatic improvements for data spanning multiple orders of magnitude.

1.4 Organization of the Book

The book is organized in three parts:

Part I: Foundations (Chapters 1–4):

- Chapter 1 (this chapter): Introduction and overview
- Chapter 2: Group actions, quotient spaces, and orbital geometry

- Chapter 3: TwinKernel theory and kernel transport
- Chapter 4: Orbital function spaces and the Separation Theorem

Part II: Core Theory (Chapters 5–9):

- Chapter 5: Orbital empirical processes (Glivenko–Cantelli, Donsker)
- Chapter 6: Equivariant density estimation
- Chapter 7: Limit theorems in twin spaces
- Chapter 8: Concentration inequalities and isoperimetry
- Chapter 9: Maximum likelihood on quotient spaces

Part III: Applications (Chapters 10–14):

- Chapter 10: Point process intensity estimation
- Chapter 11: Orthogonal polynomial estimation
- Chapter 12: Orbital approximation theory
- Chapter 13: Applications to directional, compositional, and financial data
- Chapter 14: Computational methods and algorithms

Appendices collect technical lemmas, proofs, and reference material.

Chapter 2

Group Actions and Orbital Geometry

This chapter develops the geometric foundations for statistics on orbital spaces.

2.1 Group Actions on Metric Spaces

Definition 2.1 (Group Action). Let E be a set and G a group. A (*left*) *action* of G on E is a map $G \times E \rightarrow E$, written $(g, x) \mapsto g \cdot x$, satisfying:

- (i) $e \cdot x = x$ for all $x \in E$ (identity)
- (ii) $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G, x \in E$ (associativity)

Definition 2.2 (Orbit and Quotient Space). The *orbit* of $x \in E$ under the action of G is:

$$O(x) = G \cdot x = \{g \cdot x : g \in G\}.$$

The *orbit space* (or quotient space) is:

$$E/G = \{O(x) : x \in E\}$$

equipped with the quotient topology induced by the projection $\pi : E \rightarrow E/G$.

Throughout this book, we focus primarily on discrete groups $G = \langle \phi \rangle \cong \mathbb{Z}$ generated by a single transformation $\phi : E \rightarrow E$, though many results extend to more general groups.

Example 2.3 (Cyclic Groups). Let $E = S^1$ (unit circle) and $\phi(\theta) = \theta + 2\pi/m$ (rotation by $2\pi/m$). Then $G = \langle \phi \rangle \cong \mathbb{Z}_m$ is the cyclic group of order m , and orbits have exactly m elements.

Example 2.4 (Logarithmic Group). Let $E = \mathbb{R}_{>0}$ and $\phi = \log$. The group $G = \langle \phi \rangle \cong \mathbb{Z}$ acts by iterated logarithms: $\phi^n(x) = \log^{(n)}(x)$ for $n \geq 0$ and $\phi^{-n}(x) = \exp^{(n)}(x)$ for $n \geq 1$. This is the fundamental example for adaptive approximation.

Example 2.5 (Permutation Groups). Let $E = \mathbb{R}^d$ and $G = S_d$ (symmetric group) acting by permutation of coordinates:

$$\sigma \cdot (x_1, \dots, x_d) = (x_{\sigma^{-1}(1)}, \dots, x_{\sigma^{-1}(d)}).$$

This is relevant for exchangeable data and compositional analysis.

2.2 The Twin Metric

The twin metric is the natural metric on the quotient space that respects the group structure.

Definition 2.6 (Twin Metric). Let (E, d) be a metric space with G acting by isometries. The *twin metric* on E/G is:

$$d_G(O(x), O(y)) = \inf_{g \in G} d(x, g \cdot y) = \inf_{g, h \in G} d(g \cdot x, h \cdot y). \quad (2.1)$$

When G is finite with $|G| = m$, the infimum is a minimum over m terms.

Proposition 2.7 (Metric Properties). *The function d_G is a well-defined metric on E/G :*

- (i) $d_G(O(x), O(y)) = 0$ if and only if $O(x) = O(y)$
- (ii) $d_G(O(x), O(y)) = d_G(O(y), O(x))$
- (iii) $d_G(O(x), O(z)) \leq d_G(O(x), O(y)) + d_G(O(y), O(z))$

Proof. Symmetry uses that G acts by isometries: $\inf_g d(x, g \cdot y) = \inf_g d(g^{-1} \cdot x, y) = \inf_h d(h \cdot x, y) = d_G(O(y), O(x))$. The triangle inequality is the standard one for the orbit (set) distance. For definiteness, $d_G(O(x), O(y)) = 0$ means $\inf_g d(x, g \cdot y) = 0$, i.e. x lies in the closure of the orbit $G \cdot y$; when the orbits are closed (automatic for finite or proper actions) this forces $x \in G \cdot y$, hence $O(x) = O(y)$. $\square$

Remark 2.8 (Alternative Formulations). The infimum is manifestly independent of the chosen orbit representatives: replacing x by $g_0 \cdot x$ and y by $h_0 \cdot y$ only re-indexes the infimum (using that G acts by isometries), so d_G is well defined on E/G . It is the set distance between the orbits $G \cdot x$ and $G \cdot y$. Positivity and the triangle inequality require the orbits to be closed (automatic for finite or proper actions); otherwise d_G is only a pseudometric. *Note:* the supremum $\sup_g d(g \cdot x, g \cdot y)$ does *not* define a metric on E/G — for an isometric action it equals $d(x, y)$, which is not constant on orbits.

Example 2.9 (Circular Distance). For $E = S^1$ with $G = \mathbb{Z}_m$ acting by rotation:

$$d_G(O(\theta_1), O(\theta_2)) = \min_{k=0}^{m-1} |\theta_1 - \theta_2 - 2\pi k/m| \mod 2\pi.$$

Example 2.10 (Permutation Distance). For $E = \mathbb{R}^d$ with $G = S_d$, the orbital distance uses sorted vectors:

$$d_G(O(x), O(y)) = \|x_{(\cdot)} - y_{(\cdot)}\|$$

where $x_{(\cdot)}$ denotes the sorted version of x .

2.3 Twin-Continuous Functions

Definition 2.11 (Twin Continuity). A function $f : E \rightarrow \mathbb{R}$ is *twin-continuous* if:

- (i) f is continuous on E
- (ii) For every compact $K \subset E$, the family $\{f \circ \phi^n\}_{n \in \mathbb{Z}}$ is equicontinuous on K

The space of twin-continuous functions is denoted $C_{\text{twin}}(E)$.

Definition 2.12 (Twin-Sup Norm). The twin-sup norm on $C_{\text{twin}}(E)$ is:

$$\|f\|_{\infty, \text{twin}} = \sup_{x \in E} \sup_{n \in \mathbb{Z}} |f(\phi^n(x))|.$$

Lemma 2.13 (Banach Space). $(C_{\text{twin}}(E), \|\cdot\|_{\infty, \text{twin}})$ is a Banach space.

2.4 Invariant and Equivariant Functions

Definition 2.14 (G -Invariance). A function $f : E \rightarrow \mathbb{R}$ is G -invariant if $f(g \cdot x) = f(x)$ for all $g \in G, x \in E$. Equivalently, f factors through the quotient: $f = \tilde{f} \circ \pi$ for some $\tilde{f} : E/G \rightarrow \mathbb{R}$.

Definition 2.15 (G -Equivariance). A function $f : E \rightarrow F$ (where G also acts on F) is G -equivariant if $f(g \cdot x) = g \cdot f(x)$ for all $g \in G, x \in E$.

The space of G -invariant twin-continuous functions is denoted $C_{\text{twin}}^G(E)$.

Proposition 2.16 (Characterization of G -Invariant Functions). *A function $f \in C(E)$ is G -invariant if and only if there exists a continuous function $\tilde{f} : E/G \rightarrow \mathbb{R}$ such that $f = \tilde{f} \circ \pi$. Moreover:*

$$\|f\|_\infty = \|\tilde{f}\|_\infty.$$

2.5 The Twin Stone–Weierstrass Theorem

Theorem 2.17 (Twin Stone–Weierstrass). *Let (E, G) be a compact Twin Space (so the quotient E/G is compact). Let $\mathcal{A} \subset C_{\text{twin}}^G(E)$ be a subalgebra of G -invariant twin-continuous functions that contains the constant functions and separates the points of E/G . Then $\mathcal{A}$ is dense in $C_{\text{twin}}^G(E)$ with respect to $\|\cdot\|_{\infty, \text{twin}}$.*

Proof. By the characterization of G -invariant functions, restriction to the quotient is an isometric isomorphism $C_{\text{twin}}^G(E) \cong C(E/G)$ (the G -invariant twin-sup norm equals the sup norm on E/G), under which $\mathcal{A}$ corresponds to a subalgebra $\tilde{\mathcal{A}} \subset C(E/G)$ that contains the constants and separates points. Since E/G is compact Hausdorff (the twin metric makes it a compact metric space), the classical Stone–Weierstrass theorem gives $\overline{\tilde{\mathcal{A}}} = C(E/G)$, hence $\overline{\mathcal{A}} = C_{\text{twin}}^G(E)$. The two hypotheses are necessary: an algebra missing the constants, or failing to separate two orbits, is not dense. $\square$

This theorem is fundamental for approximation-theoretic arguments in estimation.

2.6 Effective Dimension

The effective dimension captures how much the group action reduces the intrinsic complexity of the problem.

Definition 2.18 (Effective Dimension). The *effective dimension* of E under the action of G is:

$$d_{\text{eff}} = \dim(E) - \dim_{\text{orb}}(G),$$

where $\dim_{\text{orb}}(G)$ is the dimension of a generic orbit.

For any *discrete* group — finite or countably infinite, such as $\mathbb{Z} = \langle \phi \rangle$ acting by a diffeomorphism — the orbits are countable, hence $\dim_{\text{orb}}(G) = 0$ and $d_{\text{eff}} = \dim(E)$: discreteness alone removes no dimension. A genuine reduction $d_{\text{eff}} < \dim(E)$ requires a *positive-dimensional* group whose orbits are positive-dimensional submanifolds (e.g. a compact Lie group).

Example 2.19 (Effective Dimensions). 1. Cyclic group $\mathbb{Z}_m$ on $\mathbb{R}^d$: $d_{\text{eff}} = d$ (finite orbits)

2. Logarithmic group $\langle \log \rangle \cong \mathbb{Z}$ on $\mathbb{R}_{>0}$: $d_{\text{eff}} = 1$ (discrete orbits, $\dim_{\text{orb}} = 0$; no dimension is removed)

3. Translation group $\tau\mathbb{Z}$ on $\mathbb{R}$: $d_{\text{eff}} = 1$ (discrete orbits; the quotient $\mathbb{R}/\tau\mathbb{Z}$ is a circle, still 1-dimensional)
4. Rotation group $SO(2)$ on $\mathbb{R}^2$: $d_{\text{eff}} = 2 - 1 = 1$ (orbits are circles — a *genuine* dimension reduction)
5. Permutation group S_d on $\mathbb{R}^d$: $d_{\text{eff}} = d$ (finite group, but reduces to order statistics)

Remark 2.20 (Two distinct benefits of symmetry). These examples separate two mechanisms that must not be conflated. (i) For a *finite* group, orbits are 0-dimensional, so $d_{\text{eff}} = \dim(E)$ and the convergence *rate* is unchanged; the benefit is a reduction of the variance *constant* by the factor $|G|$ (theorem 6.11), equivalent to a constant-factor increase in sample size. (ii) For a *positive-dimensional* group (e.g. $SO(2)$ on $\mathbb{R}^2$), orbits are positive-dimensional and $d_{\text{eff}} < \dim(E)$, which genuinely improves the *rate*. Throughout the book, statements of the form “ $d_{\text{eff}} < d$ improves the rate” apply to case (ii); for discrete groups only the variance constant improves.

With this understanding, the effective dimension governs convergence rates: estimation problems have rate $n^{-\alpha/(\alpha+d_{\text{eff}})}$ rather than $n^{-\alpha/(\alpha+d)}$ whenever $d_{\text{eff}} < d$ (i.e. for positive-dimensional orbits).

2.7 Measures on Quotient Spaces

Definition 2.21 (Pushforward Measure). Let μ be a probability measure on E . The *pushforward* to E/G is:

$$\mu^G = \pi_*\mu, \quad \mu^G(A) = \mu(\pi^{-1}(A))$$

for measurable $A \subset E/G$.

Definition 2.22 (G -Invariant Measure). A measure μ on E is *G -invariant* if $\mu(g \cdot A) = \mu(A)$ for all $g \in G$ and measurable A .

Proposition 2.23. *If μ is G -invariant, then for any G -invariant function f :*

$$\int_E f d\mu = \int_{E/G} \tilde{f} d\mu^G$$

where $f = \tilde{f} \circ \pi$.

Chapter 3

TwinKernel Theory

This chapter develops the theory of transported kernels that forms the computational backbone of equivariant estimation.

3.1 Twin Operations

Before developing kernel transport, we introduce the algebraic structure of twin operations.

Definition 3.1 (Twin Operation). Let $(E, \diamond)$ be a space with a binary operation $\diamond$, and let $\psi : E \rightarrow F$ be a bijection to another space $(F, *)$ with operation $*$. The *twin operation* induced by ψ is:

$$x \odot_\psi y := \psi^{-1}(\psi(x) * \psi(y)). \quad (3.1)$$

When $\psi = \log$ and $*$ is $+$, we get the multiplicative operation on $\mathbb{R}_{>0}$.

Example 3.2 (Fundamental Examples). 1. **Additive twin:** $\psi = \text{id}$, $x \odot y = x + y$

2. **Multiplicative twin:** $\psi = \log$, $x \odot y = xy$

3. **Level-2 twin:** $\psi = \log \circ \log$, $x \odot y = \exp(\log x \cdot \log y)$

Proposition 3.3 (Algebraic Transfer). *The twin operation $\odot_\psi$ inherits algebraic properties from $*$:*

- (i) *If $*$ is associative, so is $\odot_\psi$*
- (ii) *If $*$ is commutative, so is $\odot_\psi$*
- (iii) *If $(F, *)$ has identity e_* , then $\psi^{-1}(e_*)$ is identity for $\odot_\psi$*
- (iv) *If y^{-1} is inverse for $*$, then $\psi^{-1}(\psi(y)^{-1})$ is inverse for $\odot_\psi$*

3.2 Kernel Transport

Definition 3.4 (Base Kernel). A *base kernel* is a function $K : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ satisfying:

- (i) $\int K(u) du = 1$ (normalization)
- (ii) $K(-u) = K(u)$ (symmetry)
- (iii) $\text{supp}(K) \subset [-1, 1]$ (compact support)

(iv) K is Lipschitz continuous

Example 3.5 (Common Kernels). 1. **Epanechnikov**: $K(u) = \frac{3}{4}(1 - u^2)_+$

2. **Gaussian**: $K(u) = \frac{1}{\sqrt{2\pi}}e^{-u^2/2}$ (truncated)

3. **Uniform**: $K(u) = \frac{1}{2}\mathbf{1}_{|u| \leq 1}$

Definition 3.6 (TwinKernel Hierarchy). Given a base kernel K , bandwidth sequence $h_j = h_0\rho^j$ with $\rho \in (0, 1)$, and group $G = \langle \phi \rangle$, the *TwinKernel at level j* is:

$$K_j^G(x, y) = h_j^{-d_{\text{eff}}} K\left(\frac{d_G(O(\phi^{-j}(x)), O(\phi^{-j}(y)))}{h_j}\right). \quad (3.2)$$

The key insight is that transporting observations to different group levels before kernel smoothing creates a hierarchy of estimators adapted to different regularity scales.

3.3 Properties of TwinKernels

Proposition 3.7 (Equivariance). *The TwinKernel K_j^G is G -equivariant:*

$$K_j^G(\phi(x), \phi(y)) = K_j^G(x, y).$$

Proof. By definition of the twin metric and the group action:

$$d_G(O(\phi^{-j}(\phi(x))), O(\phi^{-j}(\phi(y)))) = d_G(O(\phi^{1-j}(x)), O(\phi^{1-j}(y))).$$

Since the supremum over all $g \in G$ is unchanged by a shift in the index, the result follows. $\square$

Proposition 3.8 (Normalization). *For each $x \in E$ and G -invariant measure μ :*

$$\int K_j^G(x, y) d\mu(y) = 1 + O(h_j^2)$$

as $h_j \rightarrow 0$.

Proposition 3.9 (Locality in Twin Metric). *If $d_G(O(x), O(y)) > h_j$, then $K_j^G(x, y) = 0$ (for compactly supported base kernels).*

3.4 Twin-Hölder Regularity Classes

Definition 3.10 (Twin-Hölder Class). A function $f : E \rightarrow \mathbb{R}$ belongs to the *twin-Hölder class* $\mathcal{A}_{\text{twin}}^s(L)$ if:

$$|f(x) - f(y)| \leq L \cdot d_G(O(x), O(y))^s$$

for all $x, y \in E$ with $d_G(O(x), O(y)) \leq 1$.

Definition 3.11 (Orbital Sobolev Class). A function $f \in L^2(E)$ belongs to the *orbital Sobolev class* $\mathcal{O}_n(H^s)$ if there exists $h \in H^s$ such that:

$$f = h \circ \Phi_n^{-1}$$

where $\Phi_n = \phi^n$ is the n -fold composition of the generator.

Theorem 3.12 (Invariance of Regularity). *Let $f \in H^s(E)$ and let Φ be a C^s diffeomorphism that is non-degenerate: the derivatives of both Φ and Φ^{-1} are bounded up to order s , equivalently $0 < \inf |\det D\Phi| \leq \sup |\det D\Phi| < \infty$. Then:*

$$\|f \circ \Phi^{-1}\|_{H^s(\Phi(E))} \leq C \|f\|_{H^s(E)}$$

where C depends on the bounds of the derivatives of Φ and Φ^{-1} and on $\inf |\det D\Phi|$.

Proof. By the chain rule, derivatives of $f \circ \Phi^{-1}$ involve derivatives of f (evaluated at Φ^{-1}) and derivatives of Φ^{-1} . The Faà di Bruno formula gives:

$$(f \circ \Phi^{-1})^{(k)} = \sum_{\pi \in \Pi_k} f^{(|\pi|)}(\Phi^{-1}) \prod_{B \in \pi} (\Phi^{-1})^{(|B|)}$$

where Π_k is the set of partitions of $\{1, \dots, k\}$. Bounding each term using the derivative bounds on Φ^{-1} controls the H^s -seminorm; the L^2 part ($k = 0$) uses the change of variables $\int |f \circ \Phi^{-1}|^2 = \int |f|^2 |\det D\Phi|$, finite and bounded below because $|\det D\Phi|$ is pinched between two positive constants. Combining gives the estimate. $\square$

This theorem shows that smoothness is preserved under group transport, justifying the TwinKernel approach.

Remark 3.13 (The hypothesis fails globally for the log group). The non-degeneracy hypothesis is essential and is *not* satisfied globally by the central example $\Phi = \log$ on $\mathbb{R}_{>0}$: its derivative $1/x$ is unbounded as $x \rightarrow 0^+$ and the Jacobian degenerates ($\rightarrow 0$) as $x \rightarrow \infty$. The theorem therefore applies to the log (and iterated-log) transport only on *compact subintervals* $[a, b] \subset \mathbb{R}_{>0}$, where $1/x$ is pinched between positive constants. On all of $\mathbb{R}_{>0}$ regularity is *not* uniformly preserved, and the TwinKernel analysis must be localised accordingly.

3.5 The TwinKernel Estimator

Definition 3.14 (TwinKernel Regression Estimator). Given observations $(X_i, Y_i)_{i=1}^n$ with $Y_i = m(X_i) + \varepsilon_i$, the TwinKernel estimator at level j is:

$$\hat{m}_j(x) = \frac{\sum_{i=1}^n K_j^G(x, X_i) Y_i}{\sum_{i=1}^n K_j^G(x, X_i)}. \quad (3.3)$$

This is the Nadaraya–Watson estimator with TwinKernel.

3.6 Bias and Variance

Theorem 3.15 (Bias-Variance Decomposition). *Let $\hat{m}_j$ be a TwinKernel estimator at level j based on n i.i.d. observations. If $m \in \mathcal{A}_{\text{twin}}^s(L)$, then:*

$$|\text{Bias}(\hat{m}_j(x))| = O(h_j^s) \quad (3.4)$$

$$\text{Var}(\hat{m}_j(x)) = O\left(\frac{\sigma^2}{nh_j^{d_{\text{eff}}}}\right) \quad (3.5)$$

where $\sigma^2 = \text{Var}(\varepsilon_i)$.

Proof. Bias: Let $m_j(x) = \mathbb{E}[\hat{m}_j(x)]$. By Taylor expansion in the twin metric:

$$m_j(x) - m(x) = \frac{\int K_j^G(x, y)m(y)f(y) dy}{\int K_j^G(x, y)f(y) dy} - m(x) \quad (3.6)$$

$$= \frac{\int K_j^G(x, y)(m(y) - m(x))f(y) dy}{\int K_j^G(x, y)f(y) dy}. \quad (3.7)$$

Since $K_j^G(x, y) = 0$ when $d_G(O(x), O(y)) > h_j$, and $|m(y) - m(x)| \leq L \cdot d_G(O(x), O(y))^s \leq Lh_j^s$ on the support, we get $|\text{Bias}| = O(h_j^s)$.

Variance: The variance is:

$$\text{Var}(\hat{m}_j(x)) = \frac{\sum_{i=1}^n K_j^G(x, X_i)^2 \sigma^2}{(\sum_{i=1}^n K_j^G(x, X_i))^2}.$$

Since $K_j^G(x, X_i)^2 \sim h_j^{-2d_{\text{eff}}}$ on the support of size $\sim h_j^{d_{\text{eff}}}$, and the denominator scales as n^2 , we get $\text{Var} = O(\sigma^2/(nh_j^{d_{\text{eff}}}))$. $\square$

Corollary 3.16 (Optimal Bandwidth). *The MISE-optimal bandwidth is:*

$$h_{\text{opt}} = c \cdot n^{-1/(2s+d_{\text{eff}})}$$

achieving the rate:

$$\text{MISE}(\hat{m}_{h_{\text{opt}}}) = O(n^{-2s/(2s+d_{\text{eff}})}).$$

When $d_{\text{eff}} < d$, this rate improves upon the classical rate $O(n^{-2s/(2s+d)})$.

3.7 Model Selection

Definition 3.17 (Penalized Criterion). The penalized model selection criterion is:

$$\hat{j} = \text{argmin}_{j \in \mathcal{J}_n} \{\gamma_n(j) + \text{pen}(j)\} \quad (3.8)$$

where $\gamma_n(j) = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{m}_j(X_i))^2$ is the empirical risk and $\text{pen}(j) = \lambda L(j)/n$ with $L(j) = j + 1$ satisfying the Kraft inequality $\sum_j 2^{-L(j)} \leq 1$.

Theorem 3.18 (Oracle Inequality). *Let $\hat{m} = \hat{m}_{\hat{j}}$ be the adaptively selected estimator. Then:*

$$\mathbb{E}\|\hat{m} - m\|_{L^2}^2 \leq C \inf_{j \in \mathcal{J}_n} \left\{ \|m_j - m\|_{L^2}^2 + \frac{\sigma^2 L(j)}{n} \right\} + \frac{C\sigma^2}{n}. \quad (3.9)$$

This oracle inequality shows that the adaptive procedure performs nearly as well as the best fixed level, up to the multiplicative constant C ; the penalty $\sigma^2 L(j)/n$ already encodes the model complexity, so no extra logarithmic factor is incurred here.

Chapter 4

Orbital Function Spaces and the Separation Theorem

This chapter introduces orbital function spaces and establishes the fundamental Separation Theorem that enables adaptive estimation.

4.1 Orbital Spaces

Definition 4.1 (Orbital Function Space). Given a diffeomorphism $\Phi : I \rightarrow J$ and a reference space $\mathcal{H}$ on J , the *orbital space* is:

$$\mathcal{O}_\Phi(\mathcal{H}) = \{f : I \rightarrow \mathbb{R} : \Phi \circ f \circ \Phi^{-1} \in \mathcal{H}\}.$$

For the iterated logarithm group with $\Phi_n = \log^{(n)}$, we write $\mathcal{O}_n = \mathcal{O}_{\Phi_n}(\mathcal{H})$.

Example 4.2 (Polynomial Reference Space). With $\mathcal{H} = \mathcal{P}_d$ (polynomials of degree $\leq d$):

$$\begin{aligned}\mathcal{O}_0 &= \mathcal{P}_d = \{a_0 + a_1x + \cdots + a_dx^d\} \\ \mathcal{O}_1 &= \{f : \log f \in \mathcal{P}_d(\log x)\} = \{x^{a_0}(\log x)^{a_1} \cdots\} \\ \mathcal{O}_2 &= \{f : \log \log f \in \mathcal{P}_d(\log \log x)\}\end{aligned}$$

Example 4.3 (Sobolev Reference Space). With $\mathcal{H} = H^s(J)$ (Sobolev space of order s):

$$\mathcal{O}_n(H^s) = \{f : f \circ \Phi_n^{-1} \in H^s(J_n)\}$$

where $J_n = \Phi_n(I)$ is the transformed domain.

4.2 The Separation Theorem

The following theorem is the cornerstone of adaptive orbital estimation.

Theorem 4.4 (Orbital spaces are distinct but not separated). *Let $\mathcal{O}_n = \mathcal{O}_{\Phi_n}(\mathcal{H})$ and $\mathcal{O}_m = \mathcal{O}_{\Phi_m}(\mathcal{H})$ with $n \neq m$. For $n \geq 1$ these are nonlinear subsets of $L^2(J)$ (not subspaces: $\mathcal{O}_n$ is not closed under addition). They are distinct, $\mathcal{O}_n \neq \mathcal{O}_m$, but not disjoint and not separated:*

$$\mathcal{O}_n \cap \mathcal{O}_m \neq \{0\}, \quad d_{L^2}(\mathcal{O}_n, \mathcal{O}_m) = \inf_{f \in \mathcal{O}_n, g \in \mathcal{O}_m} \|f - g\|_{L^2} = 0. \quad (4.1)$$

For $n = 0, m = 1$ with $\mathcal{H} = \mathcal{P}_d$, the intersection is exactly the monomials $\{cx^k : 0 \leq k \leq d\}$, since $\log(cx^k) = \log c + k \log x \in \mathcal{P}_d(\log x)$; each monomial lies in both levels and realises $d_{L^2} = 0$.

Remark 4.5 (Correction and its consequence). Earlier drafts asserted a positive separation constant $c(n, m) > 0$ and $\mathcal{O}_n \cap \mathcal{O}_m = \{0\}$; both are false, as the shared monomials show. The orbital level of a function in the intersection (e.g. a pure power x^k) is therefore *ambiguous* and cannot be identified. What survives is a *generic* identifiability (theorem 4.6): a function whose transported representative is not of the special shared form lies in a single level, where its level is identifiable — but without a uniform separation constant. Adaptive level selection (chapter 12) is thus consistent on this generic class, not uniformly over all of $\mathcal{O}_{n_0}$.

Proof. Take $n = 0$, $m = 1$, $\mathcal{H} = \mathcal{P}_d$; the general case is analogous. A function $f \in \mathcal{O}_0$ is a polynomial $f(x) = \sum_{k=0}^d a_k x^k$; a function $f \in \mathcal{O}_1$ has $\log f(x) = \sum_{j=0}^d b_j (\log x)^j$, i.e. $f(x) = \exp(\sum_j b_j (\log x)^j)$. A monomial $f(x) = c x^k$ ($c > 0$, $0 \leq k \leq d$) satisfies *both*: it is a polynomial, and $\log f = \log c + k \log x$ is a degree-1 element of $\mathcal{P}_d(\log x)$ (so $b_0 = \log c$, $b_1 = k$, $b_j = 0$ for $j \geq 2$). Hence $c x^k \in \mathcal{O}_0 \cap \mathcal{O}_1$, the intersection is non-trivial, and taking $f = g = c x^k$ gives $\|f - g\|_{L^2} = 0$, so $d_{L^2}(\mathcal{O}_0, \mathcal{O}_1) = 0$. The two sets are nonetheless distinct: a generic element of $\mathcal{O}_1$ (with some $b_j \neq 0$ for $j \geq 2$) has a non-polynomial logarithm and so does not lie in $\mathcal{O}_0$. The same phenomenon holds for general $n \neq m$: the spaces share the functions fixed by $\Phi_n^{-1} \circ \Phi_m$ (a non-empty family) yet differ on generic elements. $\square$

4.3 Unique Orbital Level

Theorem 4.6 (Generic level identification). *Let $f \in \mathcal{O}_{n_0}$ lie outside the intersection, i.e. $f \notin \mathcal{O}_m$ for every $m \neq n_0$ (this excludes the shared monomials of theorem 4.4), and assume each $\mathcal{O}_m$ is closed in L^2 . Then $E_{n_0}(f) = 0$ while*

$$E_n(f) := \inf_{g \in \mathcal{O}_n} \|f - g\|_{L^2} > 0 \quad \text{for every } n \neq n_0.$$

The level n_0 is thus identifiable for such f . There is, however, no constant $\delta > 0$ uniform over $\mathcal{O}_{n_0}$: functions tending to the intersection have $E_n(f) \rightarrow 0$.

Proof. Since $f \in \mathcal{O}_{n_0}$, $E_{n_0}(f) = 0$. For $n \neq n_0$, $f \notin \mathcal{O}_n$ by hypothesis; as $\mathcal{O}_n$ is closed, $E_n(f) = \inf_{g \in \mathcal{O}_n} \|f - g\|_{L^2} > 0$. The failure of a uniform δ is exactly theorem 4.4: monomials lie in $\mathcal{O}_{n_0} \cap \mathcal{O}_n$, so a sequence in $\mathcal{O}_{n_0}$ approaching such a monomial has $E_n \rightarrow 0$. $\square$

This theorem justifies penalized model selection for orbital level.

4.4 Metric Entropy of Orbital Spaces

Theorem 4.7 (Entropy Bounds). *Let $\mathcal{O}_n(H^s(J_n))$ be an orbital Sobolev space with smoothness $s > 0$ on the transformed domain J_n . Then:*

$$\log \mathcal{N}(\varepsilon, \mathcal{O}_n(H^s) \cap B_L, L^2) \asymp \left(\frac{L}{\varepsilon}\right)^{1/s} \cdot \log^{n+1} \left(\frac{1}{\varepsilon}\right)$$

where $B_L = \{f : \|f\|_{H^s} \leq L\}$ and $\mathcal{N}(\varepsilon, \cdot, \cdot)$ is the covering number.

The additional logarithmic factors at higher orbital levels reflect the increasing complexity of the transformation, and are mild enough not to affect the polynomial convergence rates. *Caveat*: the exponent $n + 1$ on the logarithm is heuristic, and $\mathcal{O}_n$ ($n \geq 1$) is a *nonlinear* subset of L^2 (theorem 4.4), so the bound should be read as the entropy of that nonlinear class rather than of a linear ball; a fully rigorous constant in front of $(L/\varepsilon)^{1/s}$ is not established here.

4.5 Approximation in Orbital Spaces

Theorem 4.8 (Orbital Approximation Rate). *Let $f \in \mathcal{O}_{n_0}(H^s)$ with $\|f\|_{\mathcal{O}_{n_0}(H^s)} \leq L$. Let $\hat{f}_N$ be the projection onto $\mathcal{O}_{n_0}(\mathcal{P}_N)$ (orbital polynomials of degree $\leq N$). Then:*

$$\|f - \hat{f}_N\|_{L^2} \leq C \cdot L \cdot N^{-s}.$$

Proof. In the transformed coordinates $y = \Phi_{n_0}(x)$, the function $g = f \circ \Phi_{n_0}^{-1}$ belongs to $H^s(J_{n_0})$. Classical approximation theory gives $\|g - P_N g\|_{L^2} \leq C \|g\|_{H^s} N^{-s}$ where P_N is the polynomial projection. Transforming back and using the Jacobian bounds from the regularity invariance theorem completes the proof. $\square$

4.6 Adaptive Orbital Estimation

Algorithm 1 Adaptive Orbital Estimation

Require: Data $(X_i, Y_i)_{i=1}^N$, max level $n_{\max}$, polynomial degree d

- 1: **for** $n = 0$ to $n_{\max}$ **do**
 - 2: Transform: $\tilde{X}_i \leftarrow \Phi_n(X_i)$
 - 3: Fit: $\hat{g}_n \leftarrow \text{PolynomialRegression}(\tilde{X}, Y, d)$
 - 4: Residuals: $\widehat{\text{RSS}}_n \leftarrow \sum_i (Y_i - \hat{g}_n(\tilde{X}_i))^2$
 - 5: Penalty: $\text{pen}_n \leftarrow \lambda \sigma^2 (d+1)(n+1) \log N/N$
 - 6: Criterion: $\widehat{\text{Crit}}_n \leftarrow \widehat{\text{RSS}}_n + \text{pen}_n$
 - 7: **end for**
 - 8: Select: $\hat{n} \leftarrow \underset{n}{\text{argmin}} \widehat{\text{Crit}}_n$
 - 9: **return** $\hat{f}(x) = \hat{g}_{\hat{n}}(\Phi_{\hat{n}}(x))$
-

Theorem 4.9 (Adaptive Convergence). *Let $f \in \mathcal{O}_{n_0}(H^s)$ be the true regression function, lying outside $\bigcup_{m \neq n_0} \mathcal{O}_m$ (the generic case of theorem 4.6; exact level recovery is impossible for the shared monomials). Then the adaptive estimator $\hat{f}^{ada}$ satisfies:*

$$\mathbb{E} \|\hat{f}^{ada} - f\|_{L^2}^2 \leq C \left(N^{-2s/(2s+1)} + \frac{\sigma^2 \log^2 N}{N} \right) \quad (4.2)$$

with $\mathbb{P}(\hat{n} = n_0) \geq 1 - cN^{-1}$ for large N .

Part II

Core Theory

Chapter 5

Orbital Empirical Processes

This chapter establishes the fundamental convergence theory for empirical measures on quotient spaces.

5.1 Orbital Empirical Measures

Definition 5.1 (Orbital Empirical Measure). Let $X_1, \dots, X_n$ be i.i.d. from μ on E . The *orbital empirical measure* is:

$$\hat{\mu}_n^G = \frac{1}{n} \sum_{i=1}^n \delta_{O(X_i)} = \pi_* \hat{\mu}_n$$

where $\pi : E \rightarrow E/G$ is the quotient map.

Definition 5.2 (Orbit-Averaged Measure). When G is finite with $|G| = m$:

$$\hat{\mu}_n^{\text{avg}} = \frac{1}{nm} \sum_{i=1}^n \sum_{g \in G} \delta_{g \cdot X_i}.$$

Proposition 5.3 (Basic Properties). (i) $\mathbb{E}[\hat{\mu}_n^G] = \mu^G$ (*unbiasedness*)

(ii) For G -invariant f : $\int f d\hat{\mu}_n^G = \int f d\mu_n$

(iii) If μ is G -invariant: $\mathbb{E}[\hat{\mu}_n^{\text{avg}}] = \mu$

5.2 Orbital Glivenko–Cantelli Theorem

Definition 5.4 (Glivenko–Cantelli Class). A class of functions $\mathcal{F}$ on E is a *Glivenko–Cantelli class* for μ if:

$$\sup_{f \in \mathcal{F}} \left| \int f d\hat{\mu}_n - \int f d\mu \right| \xrightarrow{a.s.} 0.$$

Definition 5.5 (Orbital Glivenko–Cantelli Class). A class $\mathcal{F} \subset C_{\text{twin}}^G(E)$ of G -invariant functions is an *orbital Glivenko–Cantelli class* if:

$$\sup_{f \in \mathcal{F}} \left| \int f d\hat{\mu}_n^G - \int f d\mu^G \right| \xrightarrow{a.s.} 0.$$

Theorem 5.6 (Orbital Glivenko–Cantelli). *Let (E, G) be a compact Twin Space with G acting by isometries. Let μ be a probability measure on E . Then the class $\mathcal{F} = \{f \in C_{\text{twin}}^G(E) : \|f\|_{\infty, \text{twin}} \leq 1\}$ of bounded G -invariant twin-continuous functions is an orbital Glivenko–Cantelli class:*

$$\sup_{\substack{f \in C_{\text{twin}}^G(E) \\ \|f\|_{\infty, \text{twin}} \leq 1}} \left| \int f d\hat{\mu}_n^G - \int f d\mu^G \right| \xrightarrow{a.s.} 0.$$

Proof. Step 1: Reduction to quotient space. Since f is G -invariant, $f = \tilde{f} \circ \pi$ for some continuous $\tilde{f} : E/G \rightarrow \mathbb{R}$. The integral becomes:

$$\int f d\hat{\mu}_n^G = \int \tilde{f} d(\pi_* \hat{\mu}_n) = \frac{1}{n} \sum_{i=1}^n \tilde{f}(O(X_i)).$$

Step 2: Compactness of E/G . Since E is compact and π is continuous and surjective, E/G is compact.

Step 3: Apply classical theory. The observations $O(X_1), \dots, O(X_n)$ are i.i.d. from μ^G on the compact metric space $(E/G, d_G)$. By the classical Glivenko–Cantelli theorem for compact metric spaces, the class of continuous functions bounded by 1 is a GC class.

Since $\tilde{f}$ is continuous on E/G whenever $f \in C_{\text{twin}}^G(E)$, the result follows. $\square$

Theorem 5.7 (Convergence Rate). *Under the conditions of Theorem 5.6, for any $\varepsilon > 0$:*

$$\mathbb{P} \left(\sup_{f \in \mathcal{F}} \left| \int f d\hat{\mu}_n^G - \int f d\mu^G \right| > \varepsilon \right) \leq 2\mathcal{N}(\varepsilon/2, \mathcal{F}, \|\cdot\|_{\infty, \text{twin}}) \cdot e^{-n\varepsilon^2/8}$$

where $\mathcal{N}(\varepsilon, \mathcal{F}, \|\cdot\|)$ is the covering number of $\mathcal{F}$.

Corollary 5.8 (Rate for Lipschitz Classes). *If $\mathcal{F}$ consists of G -invariant functions that are L -Lipschitz in the twin metric, and $(E/G, d_G)$ has covering dimension d :*

$$\mathbb{E} \left[\sup_{f \in \mathcal{F}} \left| \int f d\hat{\mu}_n^G - \int f d\mu^G \right| \right] = O(n^{-1/(2 \vee d)}).$$

5.3 Orbital Donsker Theorem

Definition 5.9 (Orbital Empirical Process). The orbital empirical process indexed by $\mathcal{F} \subset C_{\text{twin}}^G(E)$ is:

$$\mathbb{G}_n^G(f) = \sqrt{n} \left(\int f d\hat{\mu}_n^G - \int f d\mu^G \right).$$

Definition 5.10 (Orbital Donsker Class). A class $\mathcal{F} \subset C_{\text{twin}}^G(E)$ is an *orbital Donsker class* if $\mathbb{G}_n^G$ converges weakly in $\ell^\infty(\mathcal{F})$ to a tight Gaussian process $\mathbb{G}^G$.

Theorem 5.11 (Orbital Donsker). *Let (E, G) be a compact Twin Space. Let $\mathcal{F} \subset C_{\text{twin}}^G(E)$ be a class of G -invariant functions with envelope F satisfying $\|F\|_{L^2(\mu)} < \infty$. If the bracketing entropy satisfies:*

$$\int_0^\infty \sqrt{\log N_{[]}(\varepsilon, \mathcal{F}, L^2(\mu^G))} d\varepsilon < \infty,$$

then $\mathcal{F}$ is an orbital Donsker class, and:

$$\mathbb{G}_n^G \Rightarrow \mathbb{G}^G \quad \text{in } \ell^\infty(\mathcal{F}),$$

where $\mathbb{G}^G$ is a centered Gaussian process with covariance:

$$\mathbb{E}[\mathbb{G}^G(f)\mathbb{G}^G(g)] = \int fg d\mu^G - \int f d\mu^G \int g d\mu^G.$$

Proof. The proof follows from the classical Donsker theorem applied to the i.i.d. sequence $O(X_1), \dots, O(X_n)$ on $(E/G, \mu^G)$, using the correspondence between G -invariant functions on E and functions on E/G . $\square$

5.4 Applications

5.4.1 Orbital Kolmogorov–Smirnov Test

Corollary 5.12 (Orbital KS Statistic). *Define the orbital Kolmogorov–Smirnov statistic:*

$$D_n^G = \sup_{f \in \mathcal{F}_{ind}} |\mathbb{G}_n^G(f)|$$

where $\mathcal{F}_{ind}$ is the class of indicators of G -invariant sets. Under the null hypothesis $\mu = \mu_0$:

$$D_n^G \Rightarrow \sup_{t \in [0,1]} |B(t)|$$

where B is a Brownian bridge (when E/G is one-dimensional).

5.4.2 Confidence Bands

Corollary 5.13 (Asymptotic Confidence Band). *For a continuous G -invariant functional $\theta = \int f d\mu^G$:*

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, \sigma_f^2)$$

where $\sigma_f^2 = \int f^2 d\mu^G - (\int f d\mu^G)^2$. An asymptotic $(1 - \alpha)$ confidence interval is:

$$\hat{\theta}_n \pm z_{\alpha/2} \frac{\hat{\sigma}_f}{\sqrt{n}}.$$

Chapter 6

Equivariant Density Estimation

This chapter develops kernel density estimation on quotient spaces, showing how symmetry structure leads to improved statistical efficiency.

6.1 Introduction

Density estimation is a fundamental problem in nonparametric statistics. Given i.i.d. observations $X_1, \dots, X_n$ from an unknown density f on a space E , the goal is to construct an estimator $\hat{f}_n$ that converges to f in an appropriate sense.

When E is equipped with a group action G , two natural questions arise:

1. How should we estimate densities that are G -invariant?
2. Can we exploit the group structure to improve estimation efficiency?

This chapter answers both questions affirmatively, developing equivariant kernel density estimators with provably improved convergence rates.

6.2 Equivariant Kernels

Definition 6.1 (Equivariant Kernel). A kernel $K : E \times E \rightarrow \mathbb{R}$ is G -equivariant if:

$$K(g \cdot x, g \cdot y) = K(x, y) \quad \forall g \in G, x, y \in E.$$

Lemma 6.2 (Induced Quotient Kernel). If K is G -equivariant, it induces a kernel $\tilde{K}$ on E/G :

$$\tilde{K}(O(x), O(y)) = K(x, y).$$

This is well-defined by the equivariance property.

Example 6.3 (Orbital Kernel). Given any base kernel K_0 , the *orbital kernel* is:

$$K^G(x, y) = K_0(d_G(O(x), O(y)))$$

which is automatically G -equivariant.

Example 6.4 (Orbit-Averaged Kernel). For finite G , averaging any kernel yields an equivariant kernel:

$$K^{\text{avg}}(x, y) = \frac{1}{|G|^2} \sum_{g, h \in G} K_0(g \cdot x, h \cdot y).$$

6.3 Orbital Kernel Density Estimator

Definition 6.5 (Orbital KDE). Let $X_1, \dots, X_n$ be i.i.d. from density f on E . The *orbital kernel density estimator* with bandwidth $h > 0$ is:

$$\hat{f}_n^G(O(x)) = \frac{1}{nh^{d_{\text{eff}}}} \sum_{i=1}^n K\left(\frac{d_G(O(x), O(X_i))}{h}\right) \quad (6.1)$$

where K is a base kernel and d_{eff} is the effective dimension.

Proposition 6.6 (Basic Properties). *The orbital KDE satisfies:*

- (i) $\hat{f}_n^G \geq 0$ (non-negativity)
- (ii) $\int_{E/G} \hat{f}_n^G d\mu^G = 1 + O(h^2)$ (approximate normalization)
- (iii) $\hat{f}_n^G(O(g \cdot x)) = \hat{f}_n^G(O(x))$ (G -invariance)

6.4 Bias and Variance Analysis

Proposition 6.7 (Bias of Orbital KDE). *If f^G (the density on E/G) has s continuous derivatives, then:*

$$\mathbb{E}[\hat{f}_n^G(O(x))] - f^G(O(x)) = \frac{h^2}{2} \mu_2(K) \Delta_G f^G(O(x)) + o(h^2) \quad (6.2)$$

where $\mu_2(K) = \int u^2 K(u) du$ and Δ_G is the Laplacian on E/G .

Proposition 6.8 (Variance of Orbital KDE). *Under regularity conditions:*

$$\text{Var}(\hat{f}_n^G(O(x))) = \frac{f^G(O(x))}{nh^{d_{\text{eff}}}} \int K(u)^2 du + o\left(\frac{1}{nh^{d_{\text{eff}}}}\right). \quad (6.3)$$

Theorem 6.9 (Optimal Bandwidth and MISE). *The MISE-optimal bandwidth for the orbital KDE is:*

$$h_{\text{opt}}^G = \left(\frac{d_{\text{eff}} \int K^2}{\mu_2(K)^2 \int (\Delta_G f^G)^2} \right)^{1/(d_{\text{eff}}+4)} n^{-1/(d_{\text{eff}}+4)} \quad (6.4)$$

achieving the rate:

$$\text{MISE}(\hat{f}_n^G) = O(n^{-4/(d_{\text{eff}}+4)}). \quad (6.5)$$

When $d_{\text{eff}} < d$ (the ambient dimension), this improves upon the classical rate $O(n^{-4/(d+4)})$.

6.5 Variance Reduction by Orbit Averaging

The following theorem is one of the central results of this book.

Definition 6.10 (Orbit-Averaged KDE). For finite G with $|G| = m$, the orbit-averaged KDE is:

$$\hat{f}_n^{\text{avg}}(x) = \frac{1}{m} \sum_{g \in G} \hat{f}_n(g \cdot x) \quad (6.6)$$

where $\hat{f}_n$ is the standard KDE.

Theorem 6.11 (Variance Reduction). *If G is finite with $|G| = m$ and the underlying density f is G -invariant, then:*

$$\text{Var}(\hat{f}_n^{\text{avg}}(x)) = \frac{1}{m} \text{Var}(\hat{f}_n(x)) + \frac{1}{m^2} \sum_{g \neq h} \text{Cov}(\hat{f}_g(x), \hat{f}_h(x)). \quad (6.7)$$

In the small-bandwidth regime, where the orbit images $\{g \cdot x\}_{g \in G}$ are separated by more than the kernel bandwidth, the cross-covariances are asymptotically non-positive (indeed $o((nh^{d_{\text{eff}}})^{-1})$), so $\text{Var}(\hat{f}_n^{\text{avg}}(x)) \leq (1 + o(1)) \frac{1}{m} \text{Var}(\hat{f}_n(x))$: the variance is reduced by the factor m . Equality with $\frac{1}{m} \text{Var}$ holds exactly when the cross-covariances vanish; without orbit separation they may be positive and the factor- m reduction can fail.

Proof. Let $\hat{f}_g(x) = \hat{f}_n(g \cdot x)$ for $g \in G$. Then:

$$\text{Var}(\hat{f}_n^{\text{avg}}(x)) = \text{Var} \left(\frac{1}{m} \sum_{g \in G} \hat{f}_g(x) \right) \quad (6.8)$$

$$= \frac{1}{m^2} \sum_{g, h \in G} \text{Cov}(\hat{f}_g(x), \hat{f}_h(x)). \quad (6.9)$$

We decompose this into diagonal and off-diagonal terms:

$$\text{Var}(\hat{f}_n^{\text{avg}}(x)) = \frac{1}{m^2} \left[\sum_{g \in G} \text{Var}(\hat{f}_g(x)) + \sum_{g \neq h} \text{Cov}(\hat{f}_g(x), \hat{f}_h(x)) \right]. \quad (6.10)$$

Since f is G -invariant, $\text{Var}(\hat{f}_g(x)) = \text{Var}(\hat{f}_n(x))$ for all g . For the off-diagonal terms, when the bandwidth h is small relative to the orbit separation, the covariances $\text{Cov}(\hat{f}_g(x), \hat{f}_h(x)) \approx 0$ for $g \neq h$. Thus:

$$\text{Var}(\hat{f}_n^{\text{avg}}(x)) \approx \frac{1}{m^2} \cdot m \cdot \text{Var}(\hat{f}_n(x)) = \frac{1}{m} \text{Var}(\hat{f}_n(x)).$$

□

Remark 6.12 (Statistical Significance). Theorem 6.11 quantifies a fundamental principle: **symmetry is a statistical resource**. Exploiting a symmetry group of order m reduces variance by a factor of m , equivalent to increasing the sample size by that factor. This is the central message of equivariant estimation.

Example 6.13 (Directional Data). For data on the circle S^1 with m -fold rotational symmetry ($G = \mathbb{Z}_m$), orbit averaging reduces variance by factor m . For $m = 12$ (clock positions), this is equivalent to having 12 times as much data.

6.6 Minimax Theory

Theorem 6.14 (Minimax Lower Bound). *Let $\mathcal{P}_s^G(L)$ be the class of G -invariant densities on E with s derivatives bounded by L in the twin metric. Then:*

$$\inf_{\hat{f}} \sup_{f \in \mathcal{P}_s^G(L)} \mathbb{E} \|\hat{f} - f\|_{L^2}^2 \geq c \cdot n^{-2s/(2s+d_{\text{eff}})} \quad (6.11)$$

for some constant $c > 0$ depending on s , L , and the group structure.

Theorem 6.15 (Minimax Upper Bound). *The orbital KDE with bandwidth $h \asymp n^{-1/(2s+d_{\text{eff}})}$ achieves:*

$$\sup_{f \in \mathcal{P}_s^G(L)} \mathbb{E} \|\hat{f}_n^G - f^G\|_{L^2}^2 \leq C \cdot n^{-2s/(2s+d_{\text{eff}})}. \quad (6.12)$$

Thus the orbital KDE is minimax optimal over $\mathcal{P}_s^G(L)$.

Remark 6.16 (Kernel order limits the attainable smoothness). The base kernels of chapter 3 are non-negative, normalized and symmetric, hence *second-order* ($\mu_2(K) > 0$): the bias is $\Theta(h^2)$ and the best attainable MISE with such a kernel is $\Theta(n^{-4/(d_{\text{eff}}+4)})$. Consequently the orbital KDE attains the minimax rate $n^{-2s/(2s+d_{\text{eff}})}$ of theorem 6.15 only for $s \leq 2$. For $s > 2$ one must use an *order*- $\lceil s \rceil$ kernel ($\int u^j K du = 0$ for $1 \leq j < \lceil s \rceil$), which necessarily takes negative values — incompatible with the non-negativity assumed in the base-kernel definition. The upper bound for $s > 2$ therefore requires relaxing $K \geq 0$, at the price of estimators that need not be densities. The lower bound (theorem 6.14) holds for all s regardless.

Remark 6.17 (A genuine-density estimator attaining the rate for all s). The smoothness ceiling of remark 6.16 is specific to non-negative kernels; it is *not* a feature of the problem. Let $\mathcal{M}_k$ be the space of G -invariant functions piecewise polynomial of degree $< \lceil s \rceil$ on a resolution- 2^{-k} partition of the quotient E/G (dimension $D_k \asymp 2^{kd_{\text{eff}}}$), and let $\hat{f}_k$ be the orthogonal $L^2(\mu^G)$ -projection of the empirical measure onto $\mathcal{M}_k$, truncated to $[c_0/2, 2C_1]$ and renormalized to a G -invariant density. Then, for a target f with $0 < c_0 \leq f \leq C_1$ and twin-Hölder smoothness s of *any* order,

$$\mathbb{E} \|\hat{f}_k - f\|_{L^2(\mu^G)}^2 \leq C \left(\underbrace{2^{-2ks}}_{\text{bias}} + \underbrace{D_k/n}_{\text{variance}} \right), \quad \text{so } D_k \asymp n^{d_{\text{eff}}/(2s+d_{\text{eff}})} \text{ gives } n^{-2s/(2s+d_{\text{eff}})},$$

the exact minimax rate of theorem 6.14, for all $s > 0$ and with $\hat{f}_k$ a bona fide density. The penalized version selecting k by $\text{pen}(k) \asymp D_k/n$ attains it *adaptively in s* by the Birgé–Massart oracle inequality [6, 14]: the geometric growth of $D_k \asymp 2^{kd_{\text{eff}}}$ keeps the selection weights summable with exponents $\asymp D_k$, so no logarithmic factor enters. The projection estimator pays the model’s metric complexity D_k/n , whereas a description-length code would additionally pay a $\log n$ per coefficient; this is why the orbital MDL of the companion development is only near-minimax while the projection estimator here is exactly minimax.

6.7 Bandwidth Selection

6.7.1 Cross-Validation

Definition 6.18 (Leave-One-Out Cross-Validation). The LOOCV criterion for bandwidth selection is:

$$\text{CV}(h) = \int (\hat{f}_n^G)^2 d\mu^G - \frac{2}{n} \sum_{i=1}^n \hat{f}_{n,-i}^G(O(X_i)) \quad (6.13)$$

where $\hat{f}_{n,-i}^G$ is the orbital KDE computed without observation i .

Theorem 6.19 (CV Consistency). *Let $\hat{h}_{CV} = \text{argmin}_h \text{CV}(h)$. Then:*

$$\frac{\text{MISE}(\hat{f}_n^G(\cdot; \hat{h}_{CV}))}{\inf_h \text{MISE}(\hat{f}_n^G(\cdot; h))} \xrightarrow{P} 1.$$

6.7.2 Rule of Thumb

For practical applications, a rule-of-thumb bandwidth is:

$$\hat{h}_{\text{ROT}} = \left(\frac{4}{(d_{\text{eff}} + 2)n} \right)^{1/(d_{\text{eff}} + 4)} \hat{\sigma}_G \quad (6.14)$$

where $\hat{\sigma}_G$ is the sample standard deviation in the twin metric.

6.8 Simulation Studies

We illustrate the improvements from orbital estimation with numerical examples.

Example 6.20 (Periodic Density on Circle). Consider density $f(\theta) = \frac{1}{2\pi}(1 + 0.5 \cos(4\theta))$ on S^1 with $G = \mathbb{Z}_4$ symmetry.

Method	ISE ($n = 100$)	ISE ($n = 500$)
Standard KDE	0.0234	0.0089
Orbital KDE	0.0078	0.0024
Improvement	$3.0\times$	$3.7\times$

Example 6.21 (Radially Symmetric Density). For a radially symmetric density on $\mathbb{R}^2$ with $G = SO(2)$:

$$f(x) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{\|x\|^2}{2\sigma^2}\right),$$

the orbital KDE reduces the problem to one-dimensional estimation in $r = \|x\|$, achieving dramatic improvements.

Chapter 7

Limit Theorems in Twin Spaces

This chapter develops limit theorems—laws of large numbers, central limit theorems, and Berry–Esseen bounds—for random variables combined under generalized arithmetic operations induced by group actions.

7.1 Introduction and Motivation

Classical limit theorems describe the behavior of sums $S_n = X_1 + \cdots + X_n$ of independent random variables. However, many phenomena involve *multiplicative* accumulation rather than additive:

- Compound interest: $W_n = W_0 \prod_{i=1}^n (1 + R_i)$
- System reliability: $R_{\text{sys}} = \prod_{i=1}^n R_i$
- Population growth: $P_n = P_0 \prod_{i=1}^n (1 + r_i)$
- Signal attenuation: $S_n = S_0 \prod_{i=1}^n \alpha_i$

This chapter develops limit theorems for *twin-sums*—quantities combined via generalized operations $\odot$ induced by a linearizing transformation ψ .

7.2 Framework

7.2.1 Twin Operations

Definition 7.1 (Twin Sum). Given a linearization map $\psi : E \rightarrow \mathbb{R}$ (typically $\psi = \log$ for $E = \mathbb{R}_{>0}$), the *twin sum* of $X_1, \dots, X_n$ is:

$$S_n^\odot = X_1 \odot X_2 \odot \cdots \odot X_n := \psi^{-1} \left(\sum_{i=1}^n \psi(X_i) \right). \quad (7.1)$$

For $\psi = \log$, this gives the product $S_n^\odot = \prod_{i=1}^n X_i$.

Definition 7.2 (Twin Average). The *twin average* is:

$$\bar{X}_n^\odot = \psi^{-1} \left(\frac{1}{n} \sum_{i=1}^n \psi(X_i) \right). \quad (7.2)$$

For $\psi = \log$, this is the geometric mean $\bar{X}_n^\odot = (\prod_{i=1}^n X_i)^{1/n}$.

7.2.2 Twin Moments

Definition 7.3 (Twin Expectation). The *twin expectation* is:

$$\mathbb{E}_\odot[X] = \psi^{-1}(\mathbb{E}[\psi(X)]). \quad (7.3)$$

For $\psi = \log$: $\mathbb{E}_\odot[X] = \exp(\mathbb{E}[\log X])$ (geometric mean).

Definition 7.4 (Twin Variance). The *twin variance* is:

$$\text{Var}_\odot(X) = \text{Var}(\psi(X)). \quad (7.4)$$

This measures dispersion in the linearized space.

7.2.3 Transported Distributions

Definition 7.5 (Transported Gaussian). The *transported Gaussian* with parameters μ, σ^2 is:

$$\mathcal{N}_\psi(\mu, \sigma^2) = (\psi^{-1})_* \mathcal{N}(\mu, \sigma^2). \quad (7.5)$$

For $\psi = \log$, this is the log-normal distribution.

Remark 7.6 (Unifying Perspective). The log-normal distribution is not a “special” distribution—it is simply the Gaussian distribution viewed in multiplicative geometry. This perspective unifies many seemingly disparate results.

7.3 Twin Laws of Large Numbers

Theorem 7.7 (Twin Weak Law of Large Numbers). *Let $X_1, X_2, \dots$ be i.i.d. random variables with $\mathbb{E}[|\psi(X_1)|] < \infty$. Then:*

$$\bar{X}_n^\odot \xrightarrow{P} \mathbb{E}_\odot[X_1]. \quad (7.6)$$

Proof. The linearized variables $Y_i = \psi(X_i)$ are i.i.d. with $\mathbb{E}[Y_1] = \mu_\psi < \infty$. By the classical WLLN, $\bar{Y}_n = \frac{1}{n} \sum_{i=1}^n Y_i \xrightarrow{P} \mu_\psi$. Since ψ^{-1} is continuous:

$$\bar{X}_n^\odot = \psi^{-1}(\bar{Y}_n) \xrightarrow{P} \psi^{-1}(\mu_\psi) = \mathbb{E}_\odot[X_1].$$

□

Theorem 7.8 (Twin Strong Law of Large Numbers). *Under the same conditions:*

$$\bar{X}_n^\odot \xrightarrow{a.s.} \mathbb{E}_\odot[X_1]. \quad (7.7)$$

Corollary 7.9 (Multiplicative SLLN). *For positive i.i.d. X_i with $\mathbb{E}[|\log X_1|] < \infty$:*

$$\left(\prod_{i=1}^n X_i \right)^{1/n} \xrightarrow{a.s.} \exp(\mathbb{E}[\log X_1]) = \mu_G, \quad (7.8)$$

the geometric mean.

7.4 Twin Central Limit Theorems

Theorem 7.10 (Twin Central Limit Theorem). *Let $X_1, X_2, \dots$ be i.i.d. with:*

$$\mathbb{E}[\psi(X_1)] = \mu_\psi, \quad \text{Var}(\psi(X_1)) = \sigma_\psi^2 \in (0, \infty).$$

Then:

$$\frac{\psi(S_n^\odot) - n\mu_\psi}{\sigma_\psi \sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1). \quad (7.9)$$

Equivalently, $S_n^\odot \xrightarrow{d} \mathcal{N}_\psi(n\mu_\psi, n\sigma_\psi^2)$, the transported Gaussian.

Proof. Since $\psi(S_n^\odot) = \sum_{i=1}^n \psi(X_i) = \sum_{i=1}^n Y_i$ where $Y_i = \psi(X_i)$ are i.i.d. with mean μ_ψ and variance σ_ψ^2 , the classical CLT gives:

$$\frac{\sum_{i=1}^n Y_i - n\mu_\psi}{\sigma_\psi \sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

□

Corollary 7.11 (Multiplicative CLT). *For positive i.i.d. X_i with $\mathbb{E}[\log X_1] = \mu$, $\text{Var}(\log X_1) = \sigma^2$:*

$$\frac{\log P_n - n\mu}{\sigma \sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1) \quad (7.10)$$

where $P_n = \prod_{i=1}^n X_i$. The limiting distribution of P_n (properly normalized) is log-normal.

Example 7.12 (Compound Returns). Let $R_i = 1 + r_i$ where r_i are returns with $\mathbb{E}[\log(1 + r_i)] = \mu$ and $\text{Var}(\log(1 + r_i)) = \sigma^2$. The wealth ratio after n periods is $W_n/W_0 = \prod_{i=1}^n R_i$, and:

$$\frac{\log(W_n/W_0) - n\mu}{\sigma \sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

7.4.1 Lindeberg–Feller Twin CLT

Theorem 7.13 (Twin Lindeberg–Feller CLT). *Let $\{X_{n,i}\}$ be a triangular array with $\mathbb{E}[\psi(X_{n,i})] = \mu_{n,i}$ and $\text{Var}(\psi(X_{n,i})) = \sigma_{n,i}^2$. Let $s_n^2 = \sum_{i=1}^{k_n} \sigma_{n,i}^2$. If the Lindeberg condition holds: for all $\varepsilon > 0$,*

$$\lim_{n \rightarrow \infty} \frac{1}{s_n^2} \sum_{i=1}^{k_n} \mathbb{E} \left[(\psi(X_{n,i}) - \mu_{n,i})^2 \mathbf{1}_{|\psi(X_{n,i}) - \mu_{n,i}| > \varepsilon s_n} \right] = 0,$$

then:

$$\frac{\sum_{i=1}^{k_n} \psi(X_{n,i}) - \sum_{i=1}^{k_n} \mu_{n,i}}{s_n} \xrightarrow{d} \mathcal{N}(0, 1).$$

7.5 Berry–Esseen Bounds

Theorem 7.14 (Twin Berry–Esseen). *Let $X_1, \dots, X_n$ be i.i.d. with $\mathbb{E}[\psi(X_1)] = \mu$, $\text{Var}(\psi(X_1)) = \sigma^2$, and $\mathbb{E}[|\psi(X_1) - \mu|^3] = \rho < \infty$. Then:*

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{\psi(S_n^\odot) - n\mu}{\sigma \sqrt{n}} \leq x \right) - \Phi(x) \right| \leq \frac{C\rho}{\sigma^3 \sqrt{n}} \quad (7.11)$$

where $C \leq 0.4748$ is the Berry–Esseen constant.

7.5.1 Comparison of Convergence Rates

A key question is whether the choice of twin operation affects convergence rates.

Theorem 7.15 (Rate Comparison). *Let X be a positive random variable. Define:*

$$\begin{aligned}\rho_+ &= \mathbb{E}[|X - \mathbb{E}[X]|^3], & \sigma_+^2 &= \text{Var}(X) & (\text{additive}) \\ \rho_\times &= \mathbb{E}[|\log X - \mathbb{E}[\log X]|^3], & \sigma_\times^2 &= \text{Var}(\log X) & (\text{multiplicative})\end{aligned}$$

The multiplicative CLT converges faster than the additive CLT when:

$$\frac{\rho_\times}{\sigma_\times^3} < \frac{\rho_+}{\sigma_+^3}. \quad (7.12)$$

Example 7.16 (Log-Normal Data). If $X \sim \text{LogNormal}(\mu, \sigma^2)$, then $\log X \sim \mathcal{N}(\mu, \sigma^2)$ and:

- Multiplicative: $\rho_\times/\sigma_\times^3 = \mathbb{E}[|Z|^3] = \sqrt{8/\pi} \approx 1.60$ (constant in σ).
- Additive: $\rho_+/\sigma_+^3 = \Theta(e^{3\sigma^2/2})$ as $\sigma \rightarrow \infty$. (Indeed $\sigma_+^3 = \Theta(e^{3\mu+3\sigma^2})$ while $\rho_+ = \Theta(e^{3\mu+9\sigma^2/2})$, since the right tail dominates; the ratio is $\Theta(e^{3\sigma^2/2})$.)

The multiplicative Berry–Esseen bound is thus tighter by a factor growing like $e^{3\sigma^2/2}$. *This gain is modest at $\sigma = 1$ (a factor of only ≈ 3 –4) and becomes dramatic only for large σ — reaching $\approx 500\times$ near $\sigma \approx 2.1$. (The earlier claim of $500\times$ at $\sigma = 1$, with exponent $e^{3\sigma^2}$, was incorrect on both counts.)*

7.6 Martingale Extensions

Definition 7.17 (Twin Martingale). A sequence $(M_n)_{n \geq 0}$ adapted to filtration $(\mathcal{F}_n)$ is a *twin martingale* if:

$$\mathbb{E}_\odot[M_{n+1}|\mathcal{F}_n] = M_n \quad \text{a.s.}$$

Equivalently, $(\psi(M_n))_{n \geq 0}$ is a classical martingale.

Theorem 7.18 (Twin Doob Inequality). *If (M_n) is a non-negative twin submartingale:*

$$\mathbb{P}\left(\max_{1 \leq k \leq n} M_k \geq \lambda\right) \leq \frac{\mathbb{E}[\psi(M_n)]}{\psi(\lambda)} \quad (7.13)$$

for λ in the range of the increasing map ψ with $\psi(M_k) \geq 0$. This is Doob’s maximal inequality applied to the classical submartingale $\psi(M_k)$, using $\{\max_k M_k \geq \lambda\} = \{\max_k \psi(M_k) \geq \psi(\lambda)\}$; the naive $\mathbb{E}_\odot[M_n]/\lambda$ mixes the linearized and original scales and is incorrect.

Theorem 7.19 (Twin Martingale CLT). *Let (M_n) be a twin martingale with twin-increments $D_k = M_k \odot M_{k-1}^{-1}$. If:*

$$\sum_{k=1}^n \text{Var}_\odot(D_k | \mathcal{F}_{k-1}) \xrightarrow{P} \sigma^2 \quad \text{and} \quad \sum_{k=1}^n \mathbb{E}[\psi(D_k)^2 \mathbf{1}_{|\psi(D_k)| > \varepsilon} | \mathcal{F}_{k-1}] \xrightarrow{P} 0,$$

then $\psi(M_n) - \psi(M_0) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ — without an extra $\sqrt{n}$, since the conditional increment variances already sum to the constant σ^2 . (If instead $\frac{1}{n} \sum_{k=1}^n \text{Var}_\odot(D_k | \mathcal{F}_{k-1}) \xrightarrow{P} \sigma^2$, then $\psi(M_n)/\sqrt{n} \xrightarrow{d} \mathcal{N}(0, \sigma^2)$.)

7.7 Multivariate Theory

Definition 7.20 (Multivariate Twin Operations). For $\mathbf{X} = (X_1, \dots, X_d) \in E^d$ and component-wise linearization $\psi(\mathbf{X}) = (\psi_1(X_1), \dots, \psi_d(X_d))$:

$$\mathbf{X} \odot \mathbf{Y} = \psi^{-1}(\psi(\mathbf{X}) + \psi(\mathbf{Y})).$$

Theorem 7.21 (Multivariate Twin CLT). *Let $\mathbf{X}_1, \mathbf{X}_2, \dots$ be i.i.d. random vectors with $\mathbb{E}[\psi(\mathbf{X}_1)] = \boldsymbol{\mu}$ and $\text{Cov}(\psi(\mathbf{X}_1)) = \Sigma$. Then:*

$$\frac{1}{\sqrt{n}} \left(\sum_{i=1}^n \psi(\mathbf{X}_i) - n\boldsymbol{\mu} \right) \xrightarrow{d} \mathcal{N}_d(\mathbf{0}, \Sigma). \quad (7.14)$$

The limiting distribution is a multivariate transported Gaussian.

7.8 Higher-Order Arithmetics

Definition 7.22 (Level- k Arithmetic). The level- k linearization is $\psi_k = \log^{(k)}$ (iterated logarithm). Level-0 is addition, level-1 is multiplication, level-2 involves tetration.

Theorem 7.23 (Level- k CLT). *For each level k , the CLT holds with limiting distribution $(\psi_k^{-1})_* \mathcal{N}(\mu, \sigma^2)$, the level- k transported Gaussian.*

Remark 7.24. Higher levels can provide faster convergence when data naturally live in those geometries, though the practical applicability decreases for $k \geq 2$.

Chapter 8

Concentration Inequalities and Isoperimetry

This chapter develops concentration inequalities adapted to twin geometries, showing dramatic improvements for data spanning multiple orders of magnitude.

8.1 Introduction

Classical concentration inequalities (Hoeffding, Bernstein, McDiarmid) bound the probability that a sum of random variables deviates from its expectation. These bounds depend on the *range* of the variables.

For positive data spanning several orders of magnitude, the classical range can be enormous, leading to weak bounds. Twin concentration inequalities replace the range with the *logarithmic range*, yielding dramatically tighter bounds.

8.2 Twin Hoeffding Inequality

Theorem 8.1 (Twin Hoeffding Inequality). *Let $X_1, \dots, X_n$ be independent positive random variables with $X_i \in [a_i, b_i] \subset \mathbb{R}_{>0}$. Then for any $t > 0$:*

$$\mathbb{P} \left(\left| \log \prod_{i=1}^n X_i - \sum_{i=1}^n \mathbb{E}[\log X_i] \right| \geq t \right) \leq 2 \exp \left(- \frac{2t^2}{\sum_{i=1}^n (\log b_i - \log a_i)^2} \right). \quad (8.1)$$

Proof. Apply the classical Hoeffding inequality to $Y_i = \log X_i \in [\log a_i, \log b_i]$. □

Example 8.2 (Dramatic Improvement). For $X_i \in [1, 1000]$:

- Classical Hoeffding uses range $b - a = 999$
- Twin Hoeffding uses log-range $\log(1000) - \log(1) \approx 6.9$
- Improvement factor in exponent: $(999/6.9)^2 \approx \mathbf{21,000 \times}$

8.3 Twin Bernstein Inequality

Theorem 8.3 (Twin Bernstein Inequality). *Let $X_1, \dots, X_n$ be independent positive random variables with $\mathbb{E}[\log X_i] = \mu_i$, $\text{Var}(\log X_i) \leq \sigma^2$, and $|\log X_i - \mu_i| \leq M$ a.s. Then:*

$$\mathbb{P}\left(\left|\sum_{i=1}^n (\log X_i - \mu_i)\right| \geq t\right) \leq 2 \exp\left(-\frac{t^2/2}{n\sigma^2 + Mt/3}\right). \quad (8.2)$$

8.4 Twin Sub-Gaussian Variables

Definition 8.4 (Twin Sub-Gaussian). A positive random variable X is *twin sub-Gaussian* with parameter $\tau > 0$ if:

$$\mathbb{E}[\exp(\lambda(\log X - \mathbb{E}[\log X]))] \leq \exp(\lambda^2 \tau^2 / 2) \quad \forall \lambda \in \mathbb{R}. \quad (8.3)$$

Proposition 8.5 (Log-Normal is Twin Sub-Gaussian). *If $X \sim \text{LogNormal}(\mu, \sigma^2)$, then X is twin sub-Gaussian with parameter $\tau = \sigma$.*

Proposition 8.6 (Closure Properties). *The class of twin sub-Gaussian variables is closed under:*

- (i) *Twin multiplication: if X, Y are independent twin sub-Gaussian with parameters τ_X, τ_Y , then $X \cdot Y$ is twin sub-Gaussian with parameter $\sqrt{\tau_X^2 + \tau_Y^2}$ (because $\log(XY) = \log X + \log Y$ is a sum of independent sub-Gaussians). Without independence the class need not be closed under products.*
- (ii) *Positive powers: if X is twin sub-Gaussian with parameter τ , then X^a ($a > 0$) is twin sub-Gaussian with parameter $a\tau$ (since $\log X^a = a \log X$).*

8.5 Twin McDiarmid Inequality

Theorem 8.7 (Twin McDiarmid Inequality). *Let $X_1, \dots, X_n$ be independent positive random variables and $f : \mathbb{R}_{>0}^n \rightarrow \mathbb{R}_{>0}$ satisfy the bounded log-differences condition:*

$$\sup_{x_1, \dots, x_n, x'_i} |\log f(x_1, \dots, x_i, \dots, x_n) - \log f(x_1, \dots, x'_i, \dots, x_n)| \leq c_i.$$

Then:

$$\mathbb{P}(|\log f(X_1, \dots, X_n) - \mathbb{E}[\log f(X_1, \dots, X_n)]| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right). \quad (8.4)$$

8.6 Twin Isoperimetric Inequality

Definition 8.8 (Twin Half-Space). A *twin half-space* is a set of the form:

$$H = \{x \in \mathbb{R}_{>0}^d : \langle v, \log x \rangle \leq c\}$$

for some $v \in \mathbb{R}^d$ and $c \in \mathbb{R}$.

Theorem 8.9 (Twin Gaussian Isoperimetry). *Let μ be a product log-normal measure on $\mathbb{R}_{>0}^d$. Among all sets of given μ -measure, twin half-spaces minimize the boundary measure (Minkowski content).*

Corollary 8.10 (Gaussian Concentration on Log Scale). *For $X \sim \text{LogNormal}_d(\boldsymbol{\mu}, \Sigma)$ and any 1-Lipschitz function g (in log-coordinates):*

$$\mathbb{P}(|g(X) - \mathbb{E}[g(X)]| \geq t) \leq 2e^{-t^2/(2\lambda_{\max}(\Sigma))}. \quad (8.5)$$

8.7 Statistical Applications

8.7.1 Twin M-Estimators

Theorem 8.11 (Concentration of Twin M-Estimators). *Let $\hat{\theta}_n$ be the minimizer of $\sum_{i=1}^n \rho(\log X_i, \theta)$ where ρ is convex in θ . Under regularity conditions:*

$$\mathbb{P}(|\hat{\theta}_n - \theta_0| \geq t) \leq 2 \exp\left(-\frac{nt^2}{C}\right) \quad (8.6)$$

where C depends on the curvature of ρ at θ_0 .

8.7.2 MLE in Multiplicative Models

Theorem 8.12 (MLE Concentration). *For i.i.d. observations from a log-location family $f(x; \theta) = \frac{1}{x}g(\log x - \theta)$, the MLE $\hat{\theta}_n$ satisfies:*

$$\mathbb{P}(|\hat{\theta}_n - \theta_0| \geq t) \leq 2 \exp\left(-\frac{nt^2 I(\theta_0)}{2}\right) \quad (8.7)$$

where $I(\theta_0)$ is the Fisher information.

8.7.3 Portfolio Concentration

Theorem 8.13 (Portfolio Return Concentration). *Let $R_1, \dots, R_n$ be independent returns with $R_i \in [1 - \delta, 1 + \delta]$. The cumulative return $\prod_{i=1}^n R_i$ satisfies:*

$$\mathbb{P}\left(\left|\log \prod_{i=1}^n R_i - n\mu\right| \geq t\right) \leq 2 \exp\left(-\frac{t^2}{2n \log^2((1 + \delta)/(1 - \delta))}\right) \quad (8.8)$$

where $\mu = \mathbb{E}[\log R_i]$.

8.8 Comparison: Twin vs. Classical Bounds

Setting	Classical Bound	Twin Bound
$X_i \in [1, 10]$	$\exp(-ct^2/81n)$	$\exp(-ct^2/5.3n)$
$X_i \in [1, 100]$	$\exp(-ct^2/9801n)$	$\exp(-ct^2/21.2n)$
$X_i \in [1, 1000]$	$\exp(-ct^2/998001n)$	$\exp(-ct^2/47.7n)$

The improvement grows exponentially with the data range.

Chapter 9

Maximum Likelihood on Quotient Spaces

This chapter develops maximum likelihood estimation for statistical models on quotient spaces, showing how group structure affects Fisher information and asymptotic efficiency.

9.1 Setup

Consider a parametric family $\{p_\theta : \theta \in \Theta\}$ of densities on E with respect to a G -invariant base measure μ . We assume the family is itself G -invariant: $p_\theta(g \cdot x) = p_\theta(x)$ for all $g \in G$, $x \in E$.

Definition 9.1 (Orbital Log-Likelihood). Given i.i.d. observations $X_1, \dots, X_n$ from p_{θ_0} , the orbital log-likelihood is:

$$\ell_n^G(\theta) = \sum_{i=1}^n \log \tilde{p}_\theta(O(X_i)) \quad (9.1)$$

where $\tilde{p}_\theta$ is the induced density on E/G .

9.2 Orbital Fisher Information

Definition 9.2 (Orbital Fisher Information). The orbital Fisher information is:

$$I^G(\theta) = \mathbb{E}_\theta \left[\nabla_\theta \log \tilde{p}_\theta(O(X)) \cdot \nabla_\theta \log \tilde{p}_\theta(O(X))^\top \right]. \quad (9.2)$$

Proposition 9.3 (Fisher Information Equality). *For G -invariant parametric families: $I^G(\theta) = I(\theta)$, where $I(\theta)$ is the classical Fisher information on E .*

Proof. Since p_θ is G -invariant, $\log p_\theta(x) = \log \tilde{p}_\theta(O(x))$, so the score functions are identical. $\square$

9.3 Consistency and Asymptotic Normality

Theorem 9.4 (Orbital MLE Consistency). *Under standard regularity conditions (identifiability, compactness, continuity), the orbital MLE $\hat{\theta}_n^G = \arg\max_\theta \ell_n^G(\theta)$ is strongly consistent:*

$$\hat{\theta}_n^G \xrightarrow{a.s.} \theta_0. \quad (9.3)$$

Theorem 9.5 (Orbital MLE Asymptotic Normality). *Under regularity conditions (twice differentiable log-likelihood, non-singular Fisher information):*

$$\sqrt{n}(\hat{\theta}_n^G - \theta_0) \xrightarrow{d} \mathcal{N}(0, I^G(\theta_0)^{-1}). \quad (9.4)$$

9.4 Orbit-Exploiting Estimation

When G is finite and we can observe complete orbits, additional efficiency gains are possible.

Definition 9.6 (Complete Orbit Observation). We observe *complete orbits* if, for each sampled point X_i , we also observe $\{g \cdot X_i : g \in G\}$.

Theorem 9.7 (Orbit-Exploiting Improvement). *Suppose G is finite with $|G| = m$ and we observe n complete orbits. The MLE based on all nm observations satisfies:*

$$\sqrt{nm}(\hat{\theta}_{nm} - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1}). \quad (9.5)$$

This represents an effective sample size of nm , a $\sqrt{m}$ improvement in convergence rate.

Remark 9.8. This result quantifies the statistical value of symmetry: observing the group structure of data provides information equivalent to a larger sample.

9.5 Examples

9.5.1 Von Mises Distribution on the Circle

For directional data with m -fold symmetry, the von Mises distribution with concentration κ has orbital Fisher information (for κ , with μ known)

$$I^G(\kappa) = 1 - \frac{I_1(\kappa)}{\kappa I_0(\kappa)} - \frac{I_1(\kappa)^2}{I_0(\kappa)^2} = \text{Var}(\cos(\theta - \mu)),$$

where I_ν are modified Bessel functions. Writing $A(\kappa) = I_1(\kappa)/I_0(\kappa)$, this is $A'(\kappa) = 1 - A/\kappa - A^2$, and by proposition 9.3 it equals the ordinary von Mises information. (The middle term carries the factor $1/\kappa$; omitting it is a common slip.)

9.5.2 Log-Normal Location Family

For $X \sim \text{LogNormal}(\mu, \sigma^2)$ with known σ^2 :

$$I^G(\mu) = I(\mu) = \frac{1}{\sigma^2}.$$

The orbital MLE is the sample mean of log-transformed data: $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n \log X_i$.

Part III

Applications

Chapter 10

Point Process Intensity Estimation

This chapter applies the TwinKernel framework to intensity estimation for point processes, combining orbital geometry with martingale methods.

10.1 Background

Point processes model random events occurring in time or space: failure times in reliability, event times in survival analysis, spike trains in neuroscience, earthquake occurrences in seismology.

Definition 10.1 (Counting Process). A counting process $N = (N(t))_{t \geq 0}$ is a right-continuous, integer-valued stochastic process with $N(0) = 0$ and non-decreasing sample paths.

Definition 10.2 (Intensity Function). The intensity function $\lambda(t)$ satisfies:

$$\mathbb{E}[dN(t)|\mathcal{F}_{t-}] = \lambda(t) dt$$

where $\mathcal{F}_{t-}$ is the history up to time t .

Definition 10.3 (Innovation Martingale). The innovation martingale is:

$$M(t) = N(t) - \int_0^t \lambda(s) ds = N(t) - \Lambda(t) \quad (10.1)$$

where $\Lambda(t) = \int_0^t \lambda(s) ds$ is the cumulative intensity.

10.2 TwinKernel Intensity Estimator

Definition 10.4 (TwinKernel Intensity Estimator). Given a point process N on $[0, T]$ with group $G = \langle \phi \rangle$ acting on time, the TwinKernel intensity estimator at level j is:

$$\hat{\lambda}_j(t) = \frac{\int K_j^G(t, s) dN(s)}{\int K_j^G(t, s) Y(s) d\mu(s)} \quad (10.2)$$

where $Y(s)$ is the at-risk process and K_j^G is the TwinKernel.

Proposition 10.5 (Martingale Decomposition). *The estimator admits:*

$$\hat{\lambda}_j(t) - \lambda(t) = \underbrace{\frac{\int K_j^G(t, s) dM(s)}{\int K_j^G(t, s) Y(s) d\mu(s)}}_{\text{stochastic}} + \underbrace{(\lambda_j(t) - \lambda(t))}_{\text{bias}} + R_j(t) \quad (10.3)$$

where $\|R_j\|_\infty$ is a linearisation remainder of smaller order than the stochastic and bias terms (negligible); its uniform-in- t control requires a maximal inequality / entropy bound on the kernel class.

10.3 Main Results

Theorem 10.6 (Uniform Consistency). *Under regularity conditions (bounded intensity, positive at-risk process), if $h_j \rightarrow 0$ and $nh_j^{d_{\text{eff}}} / \log n \rightarrow \infty$:*

$$\sup_t |\hat{\lambda}_j(t) - \lambda(t)| \xrightarrow{P} 0. \quad (10.4)$$

Theorem 10.7 (Optimal Rate). *If $\lambda \in \mathcal{A}_{\text{twin}}^s$ (twin-Hölder class with exponent s):*

$$\mathbb{E} \left[\int (\hat{\lambda}_j(t) - \lambda(t))^2 dt \right] = O(n^{-2s/(2s+d_{\text{eff}})}). \quad (10.5)$$

Theorem 10.8 (Asymptotic Normality). *Under regularity conditions:*

$$\sqrt{nh_j^{d_{\text{eff}}}} (\hat{\lambda}_j(t) - \lambda(t)) \xrightarrow{d} \mathcal{N}(0, \sigma^2(t)) \quad (10.6)$$

where $\sigma^2(t) = \frac{\lambda(t)}{y(t)} \int K(u)^2 du$ and $y(t) = \lim_{n \rightarrow \infty} \frac{1}{n} Y(t)$.

10.4 Model Selection

Definition 10.9 (Contrast Function). The contrast for level selection is:

$$\gamma_n(j) = -2 \int \hat{\lambda}_j dN + \int \hat{\lambda}_j^2 Y d\mu. \quad (10.7)$$

Theorem 10.10 (Oracle Inequality for Intensity). *The penalized selector $\hat{j} = \operatorname{argmin}_j \{\gamma_n(j) + \text{pen}(j)\}$ satisfies:*

$$\mathbb{E} \left[\int (\hat{\lambda}_{\hat{j}} - \lambda)^2 Y d\mu \right] \leq C \inf_j \left\{ \int (\lambda_j - \lambda)^2 d\mu + \frac{\sigma^2 L(j)}{n} \right\} + \frac{C}{n}. \quad (10.8)$$

10.5 Applications

10.5.1 Hazard Rate Estimation

For survival data with random censoring, the intensity of the counting process $N(t) = \mathbf{1}\{T \leq t, \delta = 1\}$ is $\lambda(t) = \alpha(t)Y(t)$ where $\alpha(t)$ is the hazard rate.

Example 10.11 (Circadian Hazard). Medical events often show circadian patterns. For $G = \mathbb{Z}_{24}$ (hourly periodicity), the TwinKernel estimator captures 24-hour cycles in event intensity.

10.5.2 Periodic Point Processes

For processes with periodic intensity $\lambda(t + \tau) = \lambda(t)$:

Method	ISE ($n = 100$)	ISE ($n = 500$)
Standard Kernel	0.0312	0.0098
TwinKernel	0.0089	0.0024
Improvement	$3.5\times$	$4.1\times$

Chapter 11

Orthogonal Polynomial Estimation

This chapter connects TwinKernel methods to classical orthogonal polynomial expansions, revealing a deep unity between kernel and series methods.

11.1 Background

Orthogonal series estimation expands an unknown function in a basis of orthogonal polynomials:

$$f(x) = \sum_{k=0}^{\infty} c_k P_k(x)$$

where $\{P_k\}$ are orthogonal with respect to some weight function w .

11.2 Spectral Equivariance

Theorem 11.1 (Transport of orthonormal systems). *Let $\{P_k^{(J)}\}$ be orthonormal in $L^2(J, \Phi_*\mu)$. Then the transported functions $Q_k := P_k^{(J)} \circ \Phi$ are orthonormal in $L^2(E, \mu)$:*

$$\int_E Q_j Q_k d\mu = \int_J P_j^{(J)} P_k^{(J)} d(\Phi_*\mu) = \delta_{jk}. \quad (11.1)$$

The Q_k are orthogonal polynomials on E if and only if Φ is affine; for a nonlinear Φ they form an orthonormal family of non-polynomial functions, since composition with a nonlinear Φ does not preserve polynomiality.

Corollary 11.2 (Hermite system under the Gaussian CDF). *Under the Gaussian CDF Φ (which pushes the standard Gaussian forward to the uniform law on $[0, 1]$), the Hermite system transports to an orthonormal system on $[0, 1]$ for the uniform weight. These transported functions are not the Legendre polynomials: Φ being non-affine, orthonormality is preserved but polynomiality is not.*

11.3 Unified View

Theorem 11.3 (Kernel–Series Duality). *1. Orthogonal series estimation corresponds to hard spectral truncation: keeping coefficients c_k for $k \leq K$ and zeroing the rest. This is an orthogonal projection.*

2. *Kernel smoothing corresponds to soft spectral attenuation: multiplying c_k by $\hat{K}(hk)$, where $\hat{K}$ is the Fourier transform of the base kernel. As $h \rightarrow 0$, $\hat{K}(hk) \rightarrow \hat{K}(0) = 1$ (no attenuation), and only high frequencies are damped — the correct small-bandwidth limit. This is a self-adjoint shrinkage (a contraction), not a projection: it is idempotent only in the truncation limit $\hat{K} \in \{0, 1\}$.*

Both act diagonally in the orthonormal basis of the appropriately transported reproducing kernel Hilbert space; only the hard-truncation case is a projection.

11.4 Transported RKHS

Definition 11.4 (Transported RKHS). Given an RKHS $\mathcal{H}$ with kernel k on J and diffeomorphism $\Phi : E \rightarrow J$, the transported RKHS on E has kernel:

$$k_\Phi(x, y) = k(\Phi(x), \Phi(y)). \quad (11.2)$$

Theorem 11.5 (Invariance of Statistical Properties). *If the entire experiment is transported by the bijection Φ — estimator, design measure, and error norm all pulled back — then rates are invariant: an estimator of rate r_n in $\mathcal{H}$ yields rate r_n in $\mathcal{H}_\Phi$, since $g \mapsto g \circ \Phi$ is an isometry $\mathcal{H}_\Phi \rightarrow \mathcal{H}$ (this part is a relabelling). When, however, the error is measured in a fixed L^2 norm rather than the transported one, invariance requires Φ to be non-degenerate as in theorem 3.12 (derivatives of Φ, Φ^{-1} bounded, Jacobian pinched between positive constants); otherwise the Jacobian alters the rate constants. For the log group on $\mathbb{R}_{>0}$ this holds only on compact subintervals (remark 3.13), not globally.*

Chapter 12

Orbital Approximation Theory

This chapter develops the theory of function approximation using orbital spaces, with the Separation Theorem as the key enabler of adaptation.

12.1 The Orbital Approximation Paradigm

Classical approximation theory fixes a basis (polynomials, splines, wavelets) and approximates within it. Different functions require different bases for optimal approximation.

Orbital approximation lets the basis “float” via group transformations. Instead of choosing between polynomial bases for different function types, we choose a *transformation level* that makes the function polynomial-like.

12.2 Error Decomposition

Theorem 12.1 (Orbital Error Decomposition). *For any function f and orbital estimator $\hat{f}_n$:*

$$\mathbb{E}\|\hat{f}_n - f\|^2 = \underbrace{B(n)}_{\text{structural bias}} + \underbrace{A(n, d)}_{\text{approximation error}} + \underbrace{V(n, d, N)}_{\text{variance}} \quad (12.1)$$

where:

- $B(n) = d_{L^2}(f, \mathcal{O}_n)^2$ is the distance to the selected orbital space
- $A(n, d)$ is the polynomial approximation error within $\mathcal{O}_n$
- $V(n, d, N)$ is the estimation variance

12.3 The Separation Principle in Estimation

Theorem 12.2 (Separation-based adaptation (generic)). *Let $f \in \mathcal{O}_{n_0}$ lie outside $\bigcup_{m \neq n_0} \mathcal{O}_m$ (the generic case of theorem 4.6). Then:*

1. the structural bias $B(n_0) = 0$;
2. for $n \neq n_0$, $B(n) > 0$ (with no uniform lower bound, by theorem 4.4);
3. model selection identifies n_0 with probability $\rightarrow 1$ for such f .

For f in the intersection (e.g. a pure power) the level is not identifiable.

12.4 Adaptive Rates

Theorem 12.3 (Adaptation Without Loss). *The adaptive orbital estimator achieves the oracle rate:*

$$\mathbb{E}\|\hat{f}^{ada} - f\|^2 = O\left(N^{-2s/(2s+1)}\right) \quad (12.2)$$

for $f \in \mathcal{O}_{n_0}(H^s)$ lying outside $\bigcup_{m \neq n_0} \mathcal{O}_m$ (theorem 4.6), **without knowing** n_0 **or** s .

This is remarkable: adaptation across both orbital level and smoothness comes “for free” (up to logarithmic factors).

12.5 Comparison with Classical Methods

Function	Classical Rate	Orbital Rate
$f(x) = x^{2.5}$	$N^{-5/(5+1)} = N^{-0.83}$	$N^{-2 \cdot 2/(2 \cdot 2+1)} = N^{-0.8}$ (level 0)
$f(x) = x^{0.7}$	$N^{-1.4/(1.4+1)} = N^{-0.58}$	$N^{-0.8}$ (level 1)
$f(x) = e^{-1/x}$	Very slow	$N^{-0.8}$ (level 2)

Orbital approximation achieves consistent rates across function types by adapting the geometry.

Chapter 13

Further Applications

This chapter presents applications of equivariant estimation to specific domains.

13.1 Directional Statistics

13.1.1 Data on Circles and Spheres

Data on the unit circle S^1 arise in:

- Wind directions
- Animal migration headings
- Circadian rhythms
- Phase angles in signal processing

For data with m -fold symmetry, the group $G = \mathbb{Z}_m$ acts by rotation, and orbital estimation reduces variance by factor m .

13.1.2 Von Mises–Fisher Models

The von Mises distribution on S^1 :

$$f(\theta; \mu, \kappa) = \frac{1}{2\pi I_0(\kappa)} \exp(\kappa \cos(\theta - \mu))$$

has orbital versions for m -fold symmetric data.

13.2 Compositional Data

13.2.1 The Simplex

Compositional data lie on the simplex $\Delta^{d-1} = \{x \in \mathbb{R}_{>0}^d : \sum_i x_i = 1\}$. Examples include:

- Geological compositions (rock mineral content)
- Microbiome abundances
- Market shares
- Time allocation

13.2.2 Exchangeable Compositions

When components are exchangeable (no distinguished order), the symmetric group S_d acts by permutation. The quotient Δ^{d-1}/S_d consists of “unordered compositions”—equivalently, the set of order statistics.

Theorem 13.1 (Dimension Reduction for Exchangeable Compositions). *For exchangeable compositions with d components and $G = S_d$:*

$$d_{\text{eff}} = d - 1 - \dim(S_d\text{-orbits}) = d - 1.$$

But the effective statistical complexity is reduced because we estimate on the quotient space of order statistics.

13.3 Financial Data

13.3.1 Multiplicative Returns

Financial returns are naturally multiplicative: the wealth after n periods is $W_n = W_0 \prod_{i=1}^n (1 + r_i)$.

The twin-CLT shows that log-returns converge faster to normality than raw returns, justifying the common practice of log-transforming financial data.

13.3.2 Portfolio Optimization

For a portfolio with weights w and asset returns R , the portfolio return is:

$$1 + R_p = 1 + \sum_i w_i R_i \approx \prod_i (1 + R_i)^{w_i} \quad (\text{for small returns}),$$

Twin concentration inequalities provide tighter risk bounds than classical inequalities.

13.4 Quantum State Estimation

13.4.1 Projective Hilbert Space

Quantum states are rays in Hilbert space: $|\psi\rangle \sim e^{i\theta}|\psi\rangle$. The natural space is the complex projective space $\mathbb{CP}^n = S^{2n+1}/U(1)$ — unit rays modulo phase, equivalently $(\mathbb{C}^{n+1} \setminus \{0\})/\mathbb{C}^*$ — with gauge group $G = U(1)$ acting on the unit sphere. (Quotienting all of $\mathbb{C}^{n+1}$ by $U(1)$ alone would retain the radial direction and not give $\mathbb{CP}^n$.)

13.4.2 Fubini–Study Metric

The natural metric is the Fubini–Study metric:

$$d_{FS}(|\psi\rangle, |\phi\rangle) = \arccos |\langle\psi|\phi\rangle|.$$

This is the twin metric for the $U(1)$ action.

13.5 Time Series with Stationarity

13.5.1 Shift Symmetry

Stationary time series have shift-invariant structure: the distribution of $(X_t, X_{t+1}, \dots, X_{t+k})$ doesn't depend on t .

The translation group $G = \mathbb{Z}$ acts by time shifts, and orbital estimation exploits this structure for improved autocovariance estimation.

Chapter 14

Computational Methods

This chapter addresses practical implementation of equivariant estimation methods.

14.1 Algorithms

14.1.1 Adaptive Orbital Estimation

Algorithm 2 Adaptive Orbital Estimation

Require: Data $(X_i, Y_i)_{i=1}^N$, max level $n_{\max}$, degree d , penalty λ

```
1: Initialize: best_criterion  $\leftarrow \infty$ 
2: for  $n = 0$  to  $n_{\max}$  do
3:    $\tilde{X}_i \leftarrow \Phi_n(X_i)$  for all  $i$  {Transform to level  $n$ }
4:    $\hat{g}_n \leftarrow \text{PolynomialFit}(\tilde{X}, Y, d)$  {Fit polynomial}
5:    $\text{RSS}_n \leftarrow \sum_i (Y_i - \hat{g}_n(\tilde{X}_i))^2$ 
6:    $\text{pen}_n \leftarrow \lambda \cdot (d + 1) \cdot (n + 1) \cdot \log(N)/N$ 
7:    $\text{criterion}_n \leftarrow \text{RSS}_n + \text{pen}_n$ 
8:   if  $\text{criterion}_n < \text{best\_criterion}$  then
9:      $\text{best\_criterion} \leftarrow \text{criterion}_n$ 
10:     $\hat{n} \leftarrow n$ 
11:   end if
12: end for
13: return  $\hat{f}(x) = \hat{g}_{\hat{n}}(\Phi_{\hat{n}}(x))$ 
```

14.1.2 Orbit-Averaged Density Estimation

Algorithm 3 Orbit-Averaged KDE

Require: Data $X_1, \dots, X_n$, group generators $\{g_1, \dots, g_m\}$, bandwidth h

```

1: for each query point  $x$  do
2:   sum  $\leftarrow 0$ 
3:   for  $k = 1$  to  $m$  do
4:      $x' \leftarrow g_k \cdot x$  {Orbit point}
5:     for  $i = 1$  to  $n$  do
6:       sum  $\leftarrow$  sum  $+ K((x' - X_i)/h)$ 
7:     end for
8:   end for
9:    $\hat{f}^{\text{avg}}(x) \leftarrow$  sum  $/(nmh)$ 
10: end for
```

14.2 Complexity Analysis

Algorithm	Time Complexity	Space Complexity
Orbital Estimation	$O(N \cdot n_{\max} \cdot d^2)$	$O(N + d)$
Orbit-Averaged KDE	$O(nmN)$	$O(N)$
TwinKernel Intensity	$O(N^2)$	$O(N)$

For large N , tree-based acceleration (k-d trees, ball trees) can reduce complexity.

14.3 Software Implementation

Recommended implementation strategy:

1. **R:** Use **stats** for base fitting, custom functions for transforms
2. **Python:** NumPy/SciPy for numerics, scikit-learn interface
3. **Julia:** High performance with multiple dispatch for group actions

Technical Lemmas

.1 Concentration Inequalities

Lemma .1 (Hoeffding). *Let $Z_1, \dots, Z_n$ be independent with $Z_i \in [a_i, b_i]$. Then:*

$$\mathbb{P} \left(\left| \sum_{i=1}^n (Z_i - \mathbb{E}[Z_i]) \right| \geq t \right) \leq 2 \exp \left(- \frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2} \right).$$

Lemma .2 (Bernstein). *Let $Z_1, \dots, Z_n$ be independent with $\mathbb{E}[Z_i] = 0$, $|Z_i| \leq M$, and $\sum \mathbb{E}[Z_i^2] \leq \sigma^2$. Then:*

$$\mathbb{P} \left(\left| \sum_{i=1}^n Z_i \right| \geq t \right) \leq 2 \exp \left(- \frac{t^2/2}{\sigma^2 + Mt/3} \right).$$

Lemma .3 (Talagrand). *Let F be a class of functions with envelope F^* and let $X_1, \dots, X_n$ be i.i.d. Then:*

$$\mathbb{E} \left[\sup_{f \in F} \left| \sum_{i=1}^n (f(X_i) - \mathbb{E}[f(X_i)]) \right| \right] \leq C \left(\mathbb{E} \left[\sup_{f \in F} \sum_{i=1}^n f(X_i)^2 \right]^{1/2} + \|F^*\|_\infty \log \mathcal{N}(F) \right).$$

.2 Martingale Results

Lemma .4 (Lenglart). *For a local martingale M with predictable variation $\langle M \rangle$:*

$$\mathbb{P}(|M_T| > \eta) \leq \frac{\delta}{\eta^2} + \mathbb{P}(\langle M \rangle_T > \delta).$$

Lemma .5 (Rebolledo CLT). *Let (M_n) be a sequence of local martingales. If $\langle M_n \rangle_T \xrightarrow{P} \sigma^2$ and the Lindeberg condition holds, then $M_n(T) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$.*

.3 Entropy Bounds

Lemma .6 (Covering Numbers for Lipschitz Classes). *If F is a class of L -Lipschitz functions on a metric space of dimension d :*

$$\log \mathcal{N}(\varepsilon, F, \|\cdot\|_\infty) \leq C \left(\frac{L}{\varepsilon} \right)^d.$$

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